AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Alexey Dosovitskiy[*,†], **Lucas Beyer**[*], **Alexander Kolesnikov**[*], **Dirk Weissenborn**[*], **Xiaohua Zhai**[*], **Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby**[*,†]

[*]equal technical contribution, [†]equal advising

Google Research, Brain Team

{adosovitskiy, neilhoulsby}@google.com

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ABSTRACT

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.[1]

1 INTRODUCTION

Self-attention-based architectures, in particular Transformers (Vaswani et al., 2017), have become the model of choice in natural language processing (NLP). The dominant approach is to pre-train on a large text corpus and then fine-tune on a smaller task-specific dataset (Devlin et al., 2019). Thanks to Transformers' computational efficiency and scalability, it has become possible to train models of unprecedented size, with over 100B parameters (Brown et al., 2020; Lepikhin et al., 2020). With the models and datasets growing, there is still no sign of saturating performance.

In computer vision, however, convolutional architectures remain dominant (LeCun et al., 1989; Krizhevsky et al., 2012; He et al., 2016). Inspired by NLP successes, multiple works try combining CNN-like architectures with self-attention (Wang et al., 2018; Carion et al., 2020), some replacing the convolutions entirely (Ramachandran et al., 2019; Wang et al., 2020a). The latter models, while theoretically efficient, have not yet been scaled effectively on modern hardware accelerators due to the use of specialized attention patterns. Therefore, in large-scale image recognition, classic ResNet-like architectures are still state of the art (Mahajan et al., 2018; Xie et al., 2020; Kolesnikov et al., 2020).

Inspired by the Transformer scaling successes in NLP, we experiment with applying a standard Transformer directly to images, with the fewest possible modifications. To do so, we split an image into patches and provide the sequence of linear embeddings of these patches as an input to a Transformer. Image patches are treated the same way as tokens (words) in an NLP application. We train the model on image classification in supervised fashion.

When trained on mid-sized datasets such as ImageNet without strong regularization, these models yield modest accuracies of a few percentage points below ResNets of comparable size. This seemingly discouraging outcome may be expected: Transformers lack some of the inductive biases

[1]Fine-tuning code and pre-trained models are available at https://github.com/google-research/vision_transformer

一张图像胜于16×16个词：用于大规模图像识别的Transformer

Alexey Dosovitskiy[*,†], **Lucas Beyer**[*], **Alexander Kolesnikov**[*], **Dirk Weissenborn**[*], **Xiaohua Zhai**[*], **Thomas Unterthiner, Mostafa Dehghani,Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby**[*,†]

[*]同等技术贡献， [†]同等指导贡献。Google Research，Brain Team。{adosovitskiy, neilhoulsby}@google.com

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摘要

尽管 Transformer 已成为自然语言处理任务事实上的标准架构，但它在计算机视觉中的应用仍然有限。在视觉领域，注意力机制通常与卷积网络结合使用，或用于替换卷积网络的部分组件，同时保留其整体结构。我们表明，对卷积神经网络的这种依赖并非必要：将纯 Transformer 直接应用于图像块序列，也能很好地完成图像分类。当使用大量数据预训练，再迁移到多个中小规模图像识别基准（ImageNet、CIFAR-100、VTAB 等）时，视觉 Transformer（ViT）相较于最先进的卷积网络取得了优异结果，同时训练所需的计算资源显著减少。[1]

1 引言

基于自注意力的架构，尤其是 Transformer（Vaswani 等，2017），已经成为自然语言处理（NLP）的首选模型。主流做法是在大型文本语料库上预训练，再在规模较小、针对特定任务的数据集上微调（Devlin 等，2019）。得益于 Transformer 的计算效率和可扩展性，人们已经能够训练规模前所未有、参数量超过1000亿的模型（Brown 等，2020；Lepikhin 等，2020）。随着模型和数据集不断扩大，性能仍未显现饱和的迹象。

然而，在计算机视觉中，卷积架构仍占据主导地位（LeCun 等，1989；Krizhevsky 等，2012；He 等，2016）。受到自然语言处理成功的启发，多项工作尝试将类似 CNN 的架构与自注意力结合（Wang 等，2018；Carion 等，2020），也有工作完全用自注意力替换卷积（Ramachandran 等，2019；Wang 等，2020a）。后者虽然在理论上高效，但由于使用专门设计的注意力模式，尚未在现代硬件加速器上实现有效扩展。因此，在大规模图像识别中，传统的 ResNet 类架构仍处于领先地位（Mahajan 等，2018；Xie 等，2020；Kolesnikov 等，2020）。

受到 Transformer 在 NLP 中成功扩展规模的启发，我们尝试尽可能少地修改标准 Transformer，直接将其用于图像。具体而言，将图像分成图像块，把每个图像块的线性嵌入组成序列，作为 Transformer 的输入。图像块的处理方式与 NLP 应用中的词元相同。我们采用监督方式，在图像分类任务上训练模型。

在 ImageNet 等中等规模数据集上训练、且不采用强正则化时，这些模型的准确率比相近规模的 ResNet 低几个百分点。这一看似不理想的结果并不意外：Transformer 缺少某些归纳偏置

[1]微调代码及预训练模型见 https://github.com/google-research/vision_transformer

inherent to CNNs, such as translation equivariance and locality, and therefore do not generalize well when trained on insufficient amounts of data.

However, the picture changes if the models are trained on larger datasets (14M-300M images). We find that large scale training trumps inductive bias. Our Vision Transformer (ViT) attains excellent results when pre-trained at sufficient scale and transferred to tasks with fewer datapoints. When pre-trained on the public ImageNet-21k dataset or the in-house JFT-300M dataset, ViT approaches or beats state of the art on multiple image recognition benchmarks. In particular, the best model reaches the accuracy of 88.55% on ImageNet, 90.72% on ImageNet-ReaL, 94.55% on CIFAR-100, and 77.63% on the VTAB suite of 19 tasks.

2 Related Work

Transformers were proposed by Vaswani et al. (2017) for machine translation, and have since become the state of the art method in many NLP tasks. Large Transformer-based models are often pre-trained on large corpora and then fine-tuned for the task at hand: BERT (Devlin et al., 2019) uses a denoising self-supervised pre-training task, while the GPT line of work uses language modeling as its pre-training task (Radford et al., 2018; 2019; Brown et al., 2020).

Naive application of self-attention to images would require that each pixel attends to every other pixel. With quadratic cost in the number of pixels, this does not scale to realistic input sizes. Thus, to apply Transformers in the context of image processing, several approximations have been tried in the past. Parmar et al. (2018) applied the self-attention only in local neighborhoods for each query pixel instead of globally. Such local multi-head dot-product self attention blocks can completely replace convolutions (Hu et al., 2019; Ramachandran et al., 2019; Zhao et al., 2020). In a different line of work, Sparse Transformers (Child et al., 2019) employ scalable approximations to global self-attention in order to be applicable to images. An alternative way to scale attention is to apply it in blocks of varying sizes (Weissenborn et al., 2019), in the extreme case only along individual axes (Ho et al., 2019; Wang et al., 2020a). Many of these specialized attention architectures demonstrate promising results on computer vision tasks, but require complex engineering to be implemented efficiently on hardware accelerators.

Most related to ours is the model of Cordonnier et al. (2020), which extracts patches of size 2×2 from the input image and applies full self-attention on top. This model is very similar to ViT, but our work goes further to demonstrate that large scale pre-training makes vanilla transformers competitive with (or even better than) state-of-the-art CNNs. Moreover, Cordonnier et al. (2020) use a small patch size of 2×2 pixels, which makes the model applicable only to small-resolution images, while we handle medium-resolution images as well.

There has also been a lot of interest in combining convolutional neural networks (CNNs) with forms of self-attention, e.g. by augmenting feature maps for image classification (Bello et al., 2019) or by further processing the output of a CNN using self-attention, e.g. for object detection (Hu et al., 2018; Carion et al., 2020), video processing (Wang et al., 2018; Sun et al., 2019), image classification (Wu et al., 2020), unsupervised object discovery (Locatello et al., 2020), or unified text-vision tasks (Chen et al., 2020c; Lu et al., 2019; Li et al., 2019).

Another recent related model is image GPT (iGPT) (Chen et al., 2020a), which applies Transformers to image pixels after reducing image resolution and color space. The model is trained in an unsupervised fashion as a generative model, and the resulting representation can then be fine-tuned or probed linearly for classification performance, achieving a maximal accuracy of 72% on ImageNet.

Our work adds to the increasing collection of papers that explore image recognition at larger scales than the standard ImageNet dataset. The use of additional data sources allows to achieve state-of-the-art results on standard benchmarks (Mahajan et al., 2018; Touvron et al., 2019; Xie et al., 2020). Moreover, Sun et al. (2017) study how CNN performance scales with dataset size, and Kolesnikov et al. (2020); Djolonga et al. (2020) perform an empirical exploration of CNN transfer learning from large scale datasets such as ImageNet-21k and JFT-300M. We focus on these two latter datasets as well, but train Transformers instead of ResNet-based models used in prior works.

这些 CNN 固有的归纳偏置，例如平移等变性和局部性，因此在训练数据不足时，泛化表现不佳。

然而，当模型在更大的数据集（1400万至3亿张图像）上训练时，情况会发生变化。我们发现，大规模训练能够胜过归纳偏置。在足够大的规模上预训练，再迁移到数据较少的任务时，我们的视觉 Transformer（ViT）表现优异。使用公开的 ImageNet-21k 或内部的 JFT-300M 预训练后，ViT 在多个图像识别基准上接近或超过最先进的方法。具体而言，最佳模型在 ImageNet、ImageNet-ReaL、CIFAR-100 和包含19个任务的 VTAB 上，分别达到88.55%、90.72%、94.55% 和77.63%的准确率。

2 相关工作

Transformer 最初由 Vaswani 等（2017）提出，用于机器翻译，此后成为多项 NLP 任务中的领先方法。大型 Transformer 模型通常先在大型语料上预训练，再针对具体任务微调：BERT（Devlin 等，2019）使用去噪自监督预训练任务，而 GPT 系列使用语言建模作为预训练任务（Radford 等，2018；2019；Brown 等，2020）。

直接把自注意力应用于图像，就需要让每个像素关注其他所有像素。其计算开销随像素数呈平方增长，因此无法扩展到实际使用的输入尺寸。为将 Transformer 用于图像处理，已有工作尝试了多种近似方法。Parmar 等（2018）仅在每个查询像素的局部邻域内计算自注意力，而非在全局计算。这类局部多头点积自注意力模块可以完全替代卷积（Hu 等，2019；Ramachandran 等，2019；Zhao 等，2020）。另一类工作中，Sparse Transformer（Child 等，2019）采用可扩展的全局自注意力近似，使其能够处理图像。扩大注意力规模的另一种方式，是在大小不同的块内计算注意力（Weissenborn 等，2019）；极端情况下，只沿单个坐标轴计算（Ho 等，2019；Wang 等，2020a）。这些专门设计的注意力架构在视觉任务上取得了有希望的结果，但要在硬件加速器上高效实现，往往需要复杂的工程设计。

与我们最相关的是 Cordonnier 等（2020）的模型：从输入图像提取大小为 2×2 的图像块，再计算完整自注意力。该模型与 ViT 很相似，但我们进一步证明，大规模预训练能使标准 Transformer 与最先进的 CNN 相媲美，甚至更好。此外，Cordonnier 等（2020）使用 2×2 像素的小图像块，因此只能处理低分辨率图像，而我们也能处理中等分辨率图像。

将卷积神经网络（CNN）与不同形式的自注意力结合，也受到了广泛关注。例如，可通过增强特征图进行图像分类（Bello 等，2019），或用自注意力进一步处理 CNN 的输出，以完成目标检测（Hu 等，2018；Carion 等，2020）、视频处理（Wang 等，2018；Sun 等，2019）、图像分类（Wu 等，2020）、无监督物体发现（Locatello 等，2020），以及统一的文本—视觉任务（Chen 等，2020c；Lu 等，2019；Li 等，2019）。

另一个近期相关模型是 image GPT（iGPT）（Chen 等，2020a）。它先降低图像分辨率和颜色空间，再将 Transformer 应用于图像像素。该模型以无监督方式训练成生成模型，所得表征可微调或进行线性探测，以评估分类性能；其在 ImageNet 上的最高准确率为72%。

越来越多论文开始研究规模超出标准 ImageNet 数据集的图像识别，我们的工作也属于这一方向。使用额外数据源，有助于在标准基准上取得领先结果（Mahajan 等，2018；Touvron 等，2019；Xie 等，2020）。此外，Sun 等（2017）研究了 CNN 性能如何随数据集规模变化；Kolesnikov 等（2020）和 Djolonga 等（2020）则实证考察了 CNN 从 ImageNet-21k、JFT-300M 等大规模数据集进行迁移学习的表现。我们同样关注后面这两个数据集，但训练的是 Transformer，而非以往使用的 ResNet 类模型。

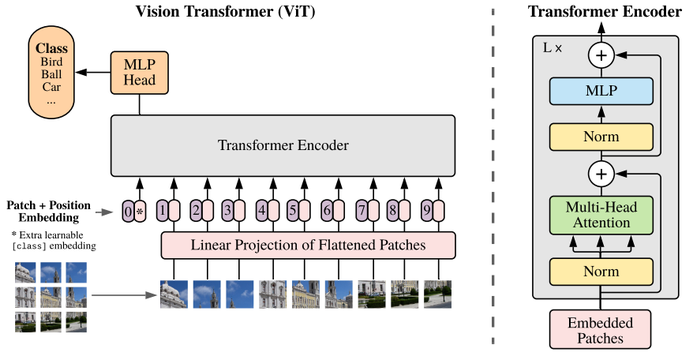


Figure 1: Model overview. We split an image into fixed-size patches, linearly embed each of them, add position embeddings, and feed the resulting sequence of vectors to a standard Transformer encoder. In order to perform classification, we use the standard approach of adding an extra learnable "classification token" to the sequence. The illustration of the Transformer encoder was inspired by Vaswani et al. (2017).

3 METHOD

In model design we follow the original Transformer (Vaswani et al., 2017) as closely as possible. An advantage of this intentionally simple setup is that scalable NLP Transformer architectures – and their efficient implementations – can be used almost out of the box.

3.1 VISION TRANSFORMER (VIT)

An overview of the model is depicted in Figure 1. The standard Transformer receives as input a 1D sequence of token embeddings. To handle 2D images, we reshape the image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ into a sequence of flattened 2D patches $\mathbf{x}_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$, where (H, W) is the resolution of the original image, C is the number of channels, (P, P) is the resolution of each image patch, and $N = HW/P^2$ is the resulting number of patches, which also serves as the effective input sequence length for the Transformer. The Transformer uses constant latent vector size D through all of its layers, so we flatten the patches and map to D dimensions with a trainable linear projection (Eq. 1). We refer to the output of this projection as the patch embeddings.

Similar to BERT's `[class]` token, we prepend a learnable embedding to the sequence of embedded patches ($\mathbf{z}_0^0 = \mathbf{x}_{\text{class}}$), whose state at the output of the Transformer encoder ($\mathbf{z}_L^0$) serves as the image representation $\mathbf{y}$ (Eq. 4). Both during pre-training and fine-tuning, a classification head is attached to $\mathbf{z}_L^0$. The classification head is implemented by a MLP with one hidden layer at pre-training time and by a single linear layer at fine-tuning time.

Position embeddings are added to the patch embeddings to retain positional information. We use standard learnable 1D position embeddings, since we have not observed significant performance gains from using more advanced 2D-aware position embeddings (Appendix D.4). The resulting sequence of embedding vectors serves as input to the encoder.

The Transformer encoder (Vaswani et al., 2017) consists of alternating layers of multiheaded self-attention (MSA, see Appendix A) and MLP blocks (Eq. 2, 3). Layernorm (LN) is applied before every block, and residual connections after every block (Wang et al., 2019; Baevski & Auli, 2019).

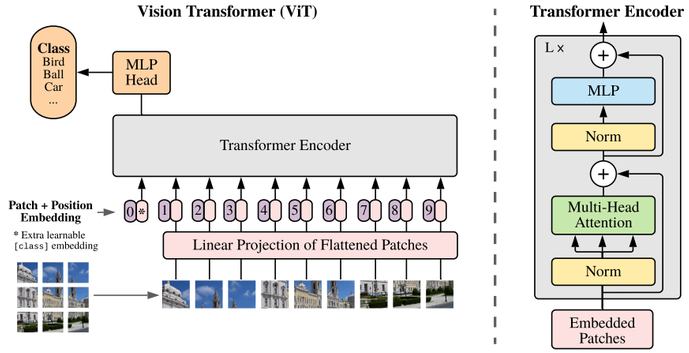


图1：模型概览。我们将图像分成固定大小的块，对每个块进行线性嵌入，加上位置嵌入，再将得到的向量序列输入标准 Transformer 编码器。分类时沿用标准做法，在序列中额外加入一个可学习的“分类词元”。Transformer 编码器示意图的设计参考了 Vaswani 等（2017）。

3 方法

模型设计尽可能遵循原始 Transformer（Vaswani 等，2017）。刻意采用简单设置的一个优势是：NLP 中可扩展的 Transformer 架构及其高效实现，几乎可以直接使用。

3.1 视觉 Transformer（ViT）

模型概览见图1。标准 Transformer 的输入是一维词元嵌入序列。为处理二维图像，我们将图像 $\mathrm{x} \in \mathbb{R}^{H \times W \times C}$ 重排为由展平的二维图像块构成的序列 $\mathrm{x}_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$，其中（$H$，$W$）是原图分辨率，$C$ 是通道数，（P，P）是每个图像块的分辨率，$N = HW/P^2$ 是所得图像块数量，同时也是 Transformer 的有效输入序列长度。Transformer 各层使用固定维度 D 的隐向量，因此我们将图像块展平，再用可训练的线性投影映射到 D 维空间（式1）。我们将该投影的输出称为图像块嵌入。

类似 BERT 的 [class] 词元，我们在图像块嵌入序列前添加一个可学习嵌入（$\mathrm{z}_0^0 = \mathrm{x}_{\text{class}}$）。它在 Transformer 编码器输出处的状态（z_L^0）作为图像表征 y（式4）。预训练和微调时，都在 z_L^0 后接一个分类头。预训练时，分类头是具有一个隐藏层的 MLP；微调时，则使用单个线性层。

我们将位置嵌入加到图像块嵌入上，以保留位置信息。使用标准的可学习一维位置嵌入，因为更复杂、显式考虑二维结构的位置嵌入，未带来明显性能提升（附录D.4）。所得嵌入向量序列构成编码器的输入。

Transformer 编码器（Vaswani 等，2017）由交替堆叠的多头自注意力（MSA，见附录A）与 MLP 块组成（式2、3）。每个块前应用层归一化（LN），每个块后使用残差连接（Wang 等，2019；Baevski 与 Auli，2019）。

The MLP contains two layers with a GELU non-linearity.

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1\mathbf{E}; \mathbf{x}_p^2\mathbf{E}; \cdots; \mathbf{x}_p^N\mathbf{E}] + \mathbf{E}_{pos}, \qquad \mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D},\ \mathbf{E}_{pos} \in \mathbb{R}^{(N+1) \times D} \tag{1}$$

$$\mathbf{z}'_\ell = \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \qquad \ell = 1 \ldots L \tag{2}$$

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, \qquad \ell = 1 \ldots L \tag{3}$$

$$\mathbf{y} = \text{LN}(\mathbf{z}_L^0) \tag{4}$$

Inductive bias. We note that Vision Transformer has much less image-specific inductive bias than CNNs. In CNNs, locality, two-dimensional neighborhood structure, and translation equivariance are baked into each layer throughout the whole model. In ViT, only MLP layers are local and translationally equivariant, while the self-attention layers are global. The two-dimensional neighborhood structure is used very sparingly: in the beginning of the model by cutting the image into patches and at fine-tuning time for adjusting the position embeddings for images of different resolution (as described below). Other than that, the position embeddings at initialization time carry no information about the 2D positions of the patches and all spatial relations between the patches have to be learned from scratch.

Hybrid Architecture. As an alternative to raw image patches, the input sequence can be formed from feature maps of a CNN (LeCun et al., 1989). In this hybrid model, the patch embedding projection $\mathbf{E}$ (Eq. 1) is applied to patches extracted from a CNN feature map. As a special case, the patches can have spatial size 1x1, which means that the input sequence is obtained by simply flattening the spatial dimensions of the feature map and projecting to the Transformer dimension. The classification input embedding and position embeddings are added as described above.

3.2 Fine-tuning and Higher Resolution

Typically, we pre-train ViT on large datasets, and fine-tune to (smaller) downstream tasks. For this, we remove the pre-trained prediction head and attach a zero-initialized $D \times K$ feedforward layer, where K is the number of downstream classes. It is often beneficial to fine-tune at higher resolution than pre-training (Touvron et al., 2019; Kolesnikov et al., 2020). When feeding images of higher resolution, we keep the patch size the same, which results in a larger effective sequence length. The Vision Transformer can handle arbitrary sequence lengths (up to memory constraints), however, the pre-trained position embeddings may no longer be meaningful. We therefore perform 2D interpolation of the pre-trained position embeddings, according to their location in the original image. Note that this resolution adjustment and patch extraction are the only points at which an inductive bias about the 2D structure of the images is manually injected into the Vision Transformer.

4 Experiments

We evaluate the representation learning capabilities of ResNet, Vision Transformer (ViT), and the hybrid. To understand the data requirements of each model, we pre-train on datasets of varying size and evaluate many benchmark tasks. When considering the computational cost of pre-training the model, ViT performs very favourably, attaining state of the art on most recognition benchmarks at a lower pre-training cost. Lastly, we perform a small experiment using self-supervision, and show that self-supervised ViT holds promise for the future.

4.1 Setup

Datasets. To explore model scalability, we use the ILSVRC-2012 ImageNet dataset with 1k classes and 1.3M images (we refer to it as ImageNet in what follows), its superset ImageNet-21k with 21k classes and 14M images (Deng et al., 2009), and JFT (Sun et al., 2017) with 18k classes and 303M high-resolution images. We de-duplicate the pre-training datasets w.r.t. the test sets of the downstream tasks following Kolesnikov et al. (2020). We transfer the models trained on these dataset to several benchmark tasks: ImageNet on the original validation labels and the cleaned-up ReaL labels (Beyer et al., 2020), CIFAR-10/100 (Krizhevsky, 2009), Oxford-IIIT Pets (Parkhi et al., 2012), and Oxford Flowers-102 (Nilsback & Zisserman, 2008). For these datasets, pre-processing follows Kolesnikov et al. (2020).

MLP 包含两个层，使用 GELU 非线性。

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1\mathbf{E}; \mathbf{x}_p^2\mathbf{E}; \cdots; \mathbf{x}_p^N\mathbf{E}] + \mathbf{E}_{pos}, \qquad \mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D},\ \mathbf{E}_{pos} \in \mathbb{R}^{(N+1) \times D} \tag{1}$$

$$\mathbf{z}'_\ell = \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \qquad \ell = 1 \ldots L \tag{2}$$

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, \qquad \ell = 1 \ldots L \tag{3}$$

$$\mathbf{y} = \text{LN}(\mathbf{z}_L^0) \tag{4}$$

归纳偏置。我们注意到，视觉 Transformer 中针对图像的归纳偏置远少于 CNN。CNN 的每一层都内置了局部性、二维邻域结构和平移等变性。在 ViT 中，只有 MLP 层具有局部性和平移等变性，而自注意力层是全局的。二维邻域结构只在极少数环节使用：模型开始时将图像切成图像块，以及微调时为不同分辨率图像调整位置嵌入，具体如下文所述。除此之外，位置嵌入在初始化时不包含图像块二维位置的信息，图像块之间的全部空间关系都必须从头学习。

混合架构。除了直接使用原始图像块，也可以从 CNN（LeCun 等，1989）的特征图构造输入序列。在这一混合模型中，图像块嵌入投影 E（式1）作用于从 CNN 特征图提取的块。特殊情况下，块的空间大小可以为1×1，即只需将特征图的空间维度展平，再投影到 Transformer 的维度，就能得到输入序列。分类输入嵌入和位置嵌入仍按前述方式添加。

3.2 微调与更高分辨率

通常，我们先在大型数据集上预训练 ViT，再在规模较小的下游任务上微调。为此，移除预训练预测头，并接入一个零初始化的 $D \times K$ 前馈层，其中 K 是下游任务的类别数。微调时使用高于预训练的分辨率通常有益（Touvron 等，2019；Kolesnikov 等，2020）。输入更高分辨率图像时，图像块大小保持不变，因此有效序列长度增大。视觉 Transformer 能处理任意长度的序列，只要内存允许；但预训练的位置嵌入可能不再适用。因此，我们依据各位置在原图中的坐标，对预训练的位置嵌入进行二维插值。请注意，分辨率调整和图像块提取，是视觉 Transformer 中仅有的两处人为注入图像二维结构归纳偏置的环节。

4 实验

我们评估 ResNet、视觉 Transformer（ViT）及混合模型的表征学习能力。为理解各模型对数据量的需求，我们在不同规模的数据集上预训练，并在多个基准任务上评估。考虑预训练计算成本时，ViT 的表现非常有利：以更低的预训练成本，在大多数识别基准上取得领先结果。最后，我们进行了一项小规模自监督实验，表明自监督 ViT 具有进一步发展的潜力。

4.1 实验设置

数据集。为研究模型的可扩展性，我们使用包含1000类、130万张图像的 ILSVRC-2012 ImageNet 数据集（下文简称 ImageNet）、其包含2.1万类、1400万张图像的超集 ImageNet-21k（Deng 等，2009），以及包含1.8万类、3.03亿张高分辨率图像的 JFT（Sun 等，2017）。按照 Kolesnikov 等（2020）的做法，我们从预训练数据中去除与下游任务测试集重复的图像。将这些数据集上训练的模型迁移到以下基准任务：使用原始验证标签及清理后的 ReaL 标签（Beyer 等，2020）的 ImageNet、CIFAR-10/100（Krizhevsky，2009）、Oxford-IIIT Pets（Parkhi 等，2012）和 Oxford Flowers-102（Nilsback 与 Zisserman，2008）。这些数据集的预处理均沿用 Kolesnikov 等（2020）的做法。

Model	Layers	Hidden size D	MLP size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

Table 1: Details of Vision Transformer model variants.

We also evaluate on the 19-task VTAB classification suite (Zhai et al., 2019b). VTAB evaluates low-data transfer to diverse tasks, using 1 000 training examples per task. The tasks are divided into three groups: *Natural* – tasks like the above, Pets, CIFAR, etc. *Specialized* – medical and satellite imagery, and *Structured* – tasks that require geometric understanding like localization.

Model Variants. We base ViT configurations on those used for BERT (Devlin et al., 2019), as summarized in Table 1. The "Base" and "Large" models are directly adopted from BERT and we add the larger "Huge" model. In what follows we use brief notation to indicate the model size and the input patch size: for instance, ViT-L/16 means the "Large" variant with 16×16 input patch size. Note that the Transformer's sequence length is inversely proportional to the square of the patch size, thus models with smaller patch size are computationally more expensive.

For the baseline CNNs, we use ResNet (He et al., 2016), but replace the Batch Normalization layers (Ioffe & Szegedy, 2015) with Group Normalization (Wu & He, 2018), and used standardized convolutions (Qiao et al., 2019). These modifications improve transfer (Kolesnikov et al., 2020), and we denote the modified model "ResNet (BiT)". For the hybrids, we feed the intermediate feature maps into ViT with patch size of one "pixel". To experiment with different sequence lengths, we either (i) take the output of stage 4 of a regular ResNet50 or (ii) remove stage 4, place the same number of layers in stage 3 (keeping the total number of layers), and take the output of this extended stage 3. Option (ii) results in a 4x longer sequence length, and a more expensive ViT model.

Training & Fine-tuning. We train all models, including ResNets, using Adam (Kingma & Ba, 2015) with $\beta_1 = 0.9$, $\beta_2 = 0.999$, a batch size of 4096 and apply a high weight decay of 0.1, which we found to be useful for transfer of all models (Appendix D.1 shows that, in contrast to common practices, Adam works slightly better than SGD for ResNets in our setting). We use a linear learning rate warmup and decay, see Appendix B.1 for details. For fine-tuning we use SGD with momentum, batch size 512, for all models, see Appendix B.1.1. For ImageNet results in Table 2, we fine-tuned at higher resolution: 512 for ViT-L/16 and 518 for ViT-H/14, and also used Polyak & Juditsky (1992) averaging with a factor of 0.9999 (Ramachandran et al., 2019; Wang et al., 2020b).

Metrics. We report results on downstream datasets either through few-shot or fine-tuning accuracy. Fine-tuning accuracies capture the performance of each model after fine-tuning it on the respective dataset. Few-shot accuracies are obtained by solving a regularized least-squares regression problem that maps the (frozen) representation of a subset of training images to $\{-1, 1\}^K$ target vectors. This formulation allows us to recover the exact solution in closed form. Though we mainly focus on fine-tuning performance, we sometimes use linear few-shot accuracies for fast on-the-fly evaluation where fine-tuning would be too costly.

4.2 Comparison to State of the Art

We first compare our largest models – ViT-H/14 and ViT-L/16 – to state-of-the-art CNNs from the literature. The first comparison point is Big Transfer (BiT) (Kolesnikov et al., 2020), which performs supervised transfer learning with large ResNets. The second is Noisy Student (Xie et al., 2020), which is a large EfficientNet trained using semi-supervised learning on ImageNet and JFT-300M with the labels removed. Currently, Noisy Student is the state of the art on ImageNet and BiT-L on the other datasets reported here. All models were trained on TPUv3 hardware, and we report the number of TPUv3-core-days taken to pre-train each of them, that is, the number of TPU v3 cores (2 per chip) used for training multiplied by the training time in days.

Table 2 shows the results. The smaller ViT-L/16 model pre-trained on JFT-300M outperforms BiT-L (which is pre-trained on the same dataset) on all tasks, while requiring substantially less computational resources to train. The larger model, ViT-H/14, further improves the performance, especially on the more challenging datasets – ImageNet, CIFAR-100, and the VTAB suite. Interestingly, this

模型	层数	隐藏维度 D	MLP 维度	头数	参数量
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

Table 1：视觉 Transformer 模型的详细配置。

我们还在包含19个任务的 VTAB 分类套件（Zhai 等，2019b）上评估。VTAB 用于评估多种任务上的少数据迁移，每个任务仅使用1000个训练样本。任务分为三组：自然类，如前述 Pets、CIFAR 等；专门领域类，如医学和卫星图像；结构类，即需要几何理解的定位等任务。

模型变体。ViT 的配置以 BERT（Devlin 等，2019）为基础，汇总见表1。"Base"和"Large"直接沿用 BERT 的配置，并新增更大的"Huge"模型。下文用简写表示模型大小和输入图像块尺寸：例如，ViT-L/16 表示输入图像块大小为 16×16 的"Large"变体。请注意，Transformer 的序列长度与图像块边长的平方成反比，因此图像块更小的模型计算开销更大。

对于卷积网络基线，我们使用 ResNet（He 等，2016），但将批归一化层（Ioffe 与 Szegedy，2015）替换为组归一化（Wu 与 He，2018），并采用标准化卷积（Qiao 等，2019）。这些修改改善了迁移效果（Kolesnikov 等，2020），我们将修改后的模型记为"ResNet（BiT）"。对于混合模型，我们把中间特征图输入 ViT，每个图像块对应一个"像素"。为考察不同序列长度，我们采用两种方案：（i）取标准 ResNet50 第4阶段的输出；（ii）移除第4阶段，将相同数量的层放入第3阶段，使总层数保持不变，再取扩展后第3阶段的输出。方案（ii）的序列长度增加到4倍，ViT 的计算开销也更大。

训练与微调。包括 ResNet 在内，所有模型都使用 Adam（Kingma 与 Ba，2015）训练，参数为 $\beta_1 = 0.9$、$\beta_2 = 0.999$，批量大小为4096，并使用较强的0.1权重衰减。我们发现，这有助于所有模型的迁移。附录 D.1表明，与常见做法不同，在我们的设置中，Adam 对 ResNet 的效果略好于 SGD。我们使用线性学习率预热与衰减，详见附录B.1。微调时，所有模型均使用带动量的 SGD，批量大小为512，详见附录B.1.1。对于表2中的 ImageNet 结果，我们采用更高分辨率微调：ViT-L/16 为512，ViT-H/14 为518；同时使用系数为0.9999的 Polyak 与 Juditsky（1992）平均（Ramachandran 等，2019；Wang 等，2020b）。

评估指标。我们在下游数据集上报告少样本准确率或微调准确率。微调准确率表示各模型在对应数据集上微调后的性能。少样本准确率则通过求解一个正则化最小二乘回归问题获得：将训练图像子集的冻结表征映射到 $\{-1, 1\}^K$ 目标向量。这一形式允许我们通过闭式表达式求得精确解。虽然主要关注微调表现，但当微调成本过高时，我们有时用线性少样本准确率进行快速的训练期间评估。

4.2 与最先进方法的比较

我们首先将最大的两个模型——ViT-H/14 和 ViT-L/16——与文献中最先进的卷积网络比较。第一个比较对象是 Big Transfer（BiT）（Kolesnikov 等，2020），它用大型 ResNet 实现监督迁移学习。第二个是 Noisy Student（Xie 等，2020），即在 ImageNet 和去掉标签的 JFT-300M 上进行半监督训练的大型 EfficientNet。目前，Noisy Student 在 ImageNet 上表现最佳，而 BiT-L 在这里报告的其他数据集上表现最佳。所有模型均在 TPUv3 硬件上训练。我们报告每个模型预训练消耗的 TPUv3 核心天数，即训练所用 TPUv3 核心数（每块芯片2个核心）与训练天数的乘积。

结果见表2。在 JFT-300M 上预训练的较小 ViT-L/16，在全部任务上优于同样在该数据集上预训练的 BiT-L，同时训练需要的计算资源显著更少。更大的 ViT-H/14 进一步提升性能，尤其在更具挑战性的 ImageNet、CIFAR-100 和 VTAB 上。有意思的是，这一

	Ours-JFT (ViT-H/14)	Ours-JFT (ViT-L/16)	Ours-I21k (ViT-L/16)	BiT-L (ResNet152x4)	Noisy Student (EfficientNet-L2)
ImageNet	**88.55** $\pm$ 0.04	87.76 $\pm$ 0.03	85.30 $\pm$ 0.02	87.54 $\pm$ 0.02	88.4/88.5*
ImageNet ReaL	**90.72** $\pm$ 0.05	90.54 $\pm$ 0.03	88.62 $\pm$ 0.05	90.54	90.55
CIFAR-10	**99.50** $\pm$ 0.06	99.42 $\pm$ 0.03	99.15 $\pm$ 0.03	99.37 $\pm$ 0.06	–
CIFAR-100	**94.55** $\pm$ 0.04	93.90 $\pm$ 0.05	93.25 $\pm$ 0.05	93.51 $\pm$ 0.08	–
Oxford-IIIT Pets	**97.56** $\pm$ 0.03	97.32 $\pm$ 0.11	94.67 $\pm$ 0.15	96.62 $\pm$ 0.23	–
Oxford Flowers-102	99.68 $\pm$ 0.02	**99.74** $\pm$ 0.00	99.61 $\pm$ 0.02	99.63 $\pm$ 0.03	–
VTAB (19 tasks)	**77.63** $\pm$ 0.23	76.28 $\pm$ 0.46	72.72 $\pm$ 0.21	76.29 $\pm$ 1.70	–
TPUv3-core-days	2.5k	0.68k	0.23k	9.9k	12.3k

Table 2: Comparison with state of the art on popular image classification benchmarks. We report mean and standard deviation of the accuracies, averaged over three fine-tuning runs. Vision Transformer models pre-trained on the JFT-300M dataset outperform ResNet-based baselines on all datasets, while taking substantially less computational resources to pre-train. ViT pre-trained on the smaller public ImageNet-21k dataset performs well too. *Slightly improved 88.5% result reported in Touvron et al. (2020).

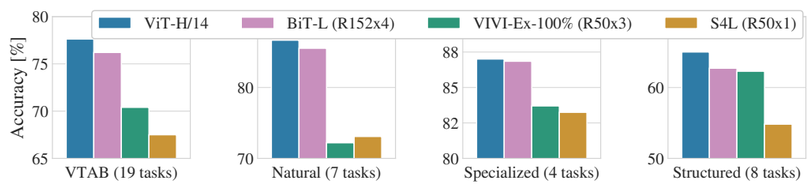


Figure 2: Breakdown of VTAB performance in *Natural*, *Specialized*, and *Structured* task groups.

model still took substantially less compute to pre-train than prior state of the art. However, we note that pre-training efficiency may be affected not only by the architecture choice, but also other parameters, such as training schedule, optimizer, weight decay, etc. We provide a controlled study of performance vs. compute for different architectures in Section 4.4. Finally, the ViT-L/16 model pre-trained on the public ImageNet-21k dataset performs well on most datasets too, while taking fewer resources to pre-train: it could be trained using a standard cloud TPUv3 with 8 cores in approximately 30 days.

Figure 2 decomposes the VTAB tasks into their respective groups, and compares to previous SOTA methods on this benchmark: BiT, VIVI – a ResNet co-trained on ImageNet and Youtube (Tschannen et al., 2020), and S4L – supervised plus semi-supervised learning on ImageNet (Zhai et al., 2019a). ViT-H/14 outperforms BiT-R152x4, and other methods, on the *Natural* and *Structured* tasks. On the *Specialized* the performance of the top two models is similar.

4.3 Pre-training Data Requirements

The Vision Transformer performs well when pre-trained on a large JFT-300M dataset. With fewer inductive biases for vision than ResNets, how crucial is the dataset size? We perform two series of experiments.

First, we pre-train ViT models on datasets of increasing size: ImageNet, ImageNet-21k, and JFT-300M. To boost the performance on the smaller datasets, we optimize three basic regularization parameters – weight decay, dropout, and label smoothing. Figure 3 shows the results after fine-tuning to ImageNet (results on other datasets are shown in Table 5)[2]. When pre-trained on the smallest dataset, ImageNet, ViT-Large models underperform compared to ViT-Base models, despite (moderate) regularization. With ImageNet-21k pre-training, their performances are similar. Only with JFT-300M, do we see the full benefit of larger models. Figure 3 also shows the performance

[2]Note that the ImageNet pre-trained models are also fine-tuned, but again on ImageNet. This is because the resolution increase during fine-tuning improves the performance.

	本文-JFT (ViT-H/14)	本文-JFT (ViT-L/16)	本文-I21k (ViT-L/16)	BiT-L (ResNet152x4)	Noisy Student (EfficientNet-L2)
ImageNet	**88.55** $\pm$ 0.04	87.76 $\pm$ 0.03	85.30 $\pm$ 0.02	87.54 $\pm$ 0.02	88.4/88.5*
ImageNet ReaL	**90.72** $\pm$ 0.05	90.54 $\pm$ 0.03	88.62 $\pm$ 0.05	90.54	90.55
CIFAR-10	**99.50** $\pm$ 0.06	99.42 $\pm$ 0.03	99.15 $\pm$ 0.03	99.37 $\pm$ 0.06	–
CIFAR-100	**94.55** $\pm$ 0.04	93.90 $\pm$ 0.05	93.25 $\pm$ 0.05	93.51 $\pm$ 0.08	–
Oxford-IIIT Pets	**97.56** $\pm$ 0.03	97.32 $\pm$ 0.11	94.67 $\pm$ 0.15	96.62 $\pm$ 0.23	–
Oxford Flowers-102	99.68 $\pm$ 0.02	**99.74** $\pm$ 0.00	99.61 $\pm$ 0.02	99.63 $\pm$ 0.03	–
VTAB（19个任务）	**77.63** $\pm$ 0.23	76.28 $\pm$ 0.46	72.72 $\pm$ 0.21	76.29 $\pm$ 1.70	–
TPUv3 核心天数	2.5k	0.68k	0.23k	9.9k	12.3k

表2：在常用图像分类基准上与最先进方法的比较。我们报告3次微调所得准确率的均值和标准差。在 JFT-300M 上预训练的视觉 Transformer，在全部数据集上优于 ResNet 基线，同时预训练所需计算资源明显更少。在较小、公开的 ImageNet-21k 上预训练的 ViT，也表现良好。 *相较 Touvron 等（2020）报告的88.5%略有提升。

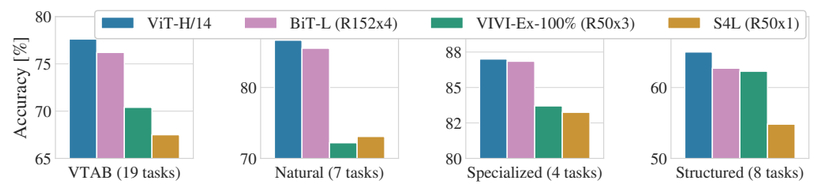


图2：性能分解 VTAB 中自然类、专门领域类 及结构类任务组。

模型的预训练计算量仍显著少于此前最先进的方法。不过，预训练效率不仅受架构选择影响，也可能受到训练安排、优化器、权重衰减等参数的影响。第4.4节对不同架构的性能与计算量进行了受控比较。最后，在公开 ImageNet-21k 上预训练的 ViT-L/16，在大多数数据集上也表现良好，同时需要更少的预训练资源：使用标准的8核云端 TPUv3，约30天即可完成训练。

图2将 VTAB 任务分成不同组，并与此前领先的方法比较：BiT、在 ImageNet 和 YouTube 上联合训练 ResNet 的 VIVI（Tschannen 等，2020），以及结合 ImageNet 监督与半监督学习的 S4L（Zhai 等，2019a）。在自然类和结构类任务上，ViT-H/14 优于 BiT-R152x4 及其他方法；在专门领域任务上，前两个模型的性能相近。

4.3 预训练的数据需求

视觉 Transformer 在大型 JFT-300M 上预训练时表现良好。与 ResNet 相比，它包含的视觉归纳偏置更少，那么数据集规模究竟有多重要？我们进行了两组实验。

首先，我们在规模逐步增大的 ImageNet、ImageNet-21k 和 JFT-300M 上预训练 ViT。为提升较小数据集上的表现，我们优化了三个基本正则化参数：权重衰减、dropout 和标签平滑。图3展示了在 ImageNet 上微调后的结果，其他数据集的结果见表5[2]。在最小的 ImageNet 上预训练时，即便采用适度正则化，ViT-Large 的表现仍不如 ViT-Base；使用 ImageNet-21k 预训练时，两者相近。只有使用 JFT-300M，我们才看到更大模型的全部优势。图3还展示了性能

[2]请注意，在 ImageNet 上预训练的模型也会微调，而且仍在 ImageNet 上。这是因为微调时提高分辨率能够改善性能。

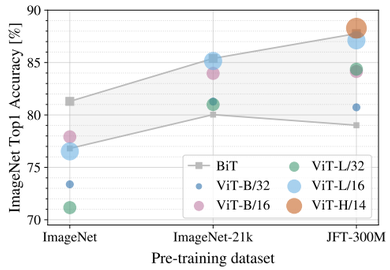


Figure 3: Transfer to ImageNet. While large ViT models perform worse than BiT ResNets (shaded area) when pre-trained on small datasets, they shine when pre-trained on larger datasets. Similarly, larger ViT variants overtake smaller ones as the dataset grows.

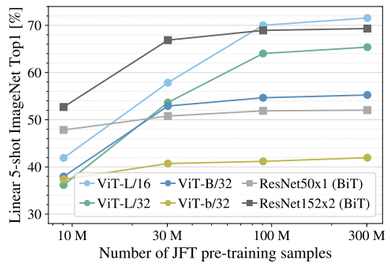


Figure 4: Linear few-shot evaluation on ImageNet versus pre-training size. ResNets perform better with smaller pre-training datasets but plateau sooner than ViT, which performs better with larger pre-training. ViT-b is ViT-B with all hidden dimensions halved.

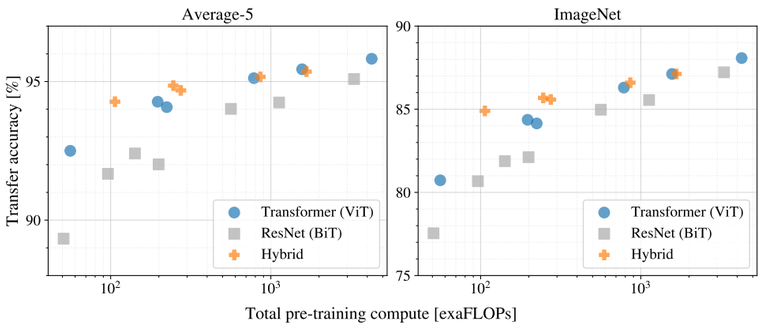


Figure 5: Performance versus pre-training compute for different architectures: Vision Transformers, ResNets, and hybrids. Vision Transformers generally outperform ResNets with the same computational budget. Hybrids improve upon pure Transformers for smaller model sizes, but the gap vanishes for larger models.

region spanned by BiT models of different sizes. The BiT CNNs outperform ViT on ImageNet, but with the larger datasets, ViT overtakes.

Second, we train our models on random subsets of 9M, 30M, and 90M as well as the full JFT-300M dataset. We do not perform additional regularization on the smaller subsets and use the same hyper-parameters for all settings. This way, we assess the intrinsic model properties, and not the effect of regularization. We do, however, use early-stopping, and report the best validation accuracy achieved during training. To save compute, we report few-shot linear accuracy instead of full fine-tuning accuracy. Figure 4 contains the results. Vision Transformers overfit more than ResNets with comparable computational cost on smaller datasets. For example, ViT-B/32 is slightly faster than ResNet50; it performs much worse on the 9M subset, but better on 90M+ subsets. The same is true for ResNet152x2 and ViT-L/16. This result reinforces the intuition that the convolutional inductive bias is useful for smaller datasets, but for larger ones, learning the relevant patterns directly from data is sufficient, even beneficial.

Overall, the few-shot results on ImageNet (Figure 4), as well as the low-data results on VTAB (Table 2) seem promising for very low-data transfer. Further analysis of few-shot properties of ViT is an exciting direction of future work.

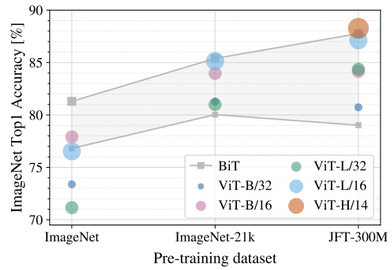


图3：迁移到 ImageNet。在小数据集上预训练时，大型 ViT 的表现不如 BiT ResNet（阴影区域）；在大数据集上预训练时，ViT 则表现出优势。同样，随着数据集扩大，较大的 ViT 变体也会超过较小的变体。

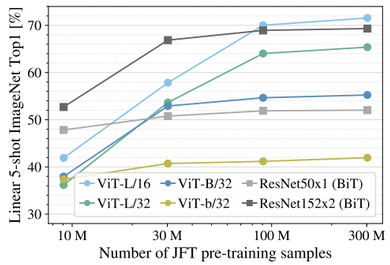


图4：ImageNet 上的线性少样本评估性能与预训练数据量之间的关系。预训练数据较少时，ResNet 表现更好，但它比 ViT 更早进入平台期；预训练数据更多时，ViT 表现更好。ViT-b 表示将 ViT-B 的全部隐藏维度减半。

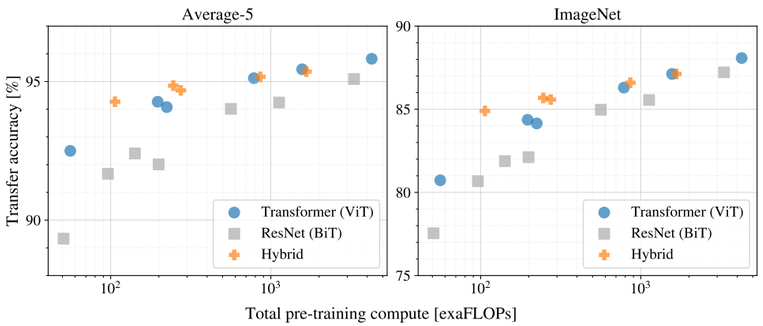


图5：视觉 Transformer、ResNet 和混合模型的性能与预训练计算量之间的关系。在相同计算预算下，视觉 Transformer 通常优于 ResNet。模型较小时，混合模型比纯 Transformer 更好；模型更大时，这一差距消失。

不同大小 BiT 模型所覆盖的性能区域。在 ImageNet 上，BiT CNN 优于 ViT；但数据集更大时，ViT 会超过它们。

其次，我们分别在随机抽取的900万、3000万、9000万张图像，以及完整的 JFT-300M 数据集上训练模型。对较小的数据子集，我们不添加额外正则化，并在所有设置中使用相同超参数，以考察模型的内在特性，而非正则化的作用。不过，我们使用早停，并报告训练期间达到的最佳验证准确率。为节省计算，我们报告少样本线性评估准确率，而非完整微调准确率。结果见图4。在较小的数据集上，视觉 Transformer 比计算开销相近的 ResNet 更容易过拟合。例如，ViT-B/32 比 ResNet50 略快，在900万图像子集上表现差得多，但在9000万+ 图像子集上表现更好。ResNet152x2 与 ViT-L/16 的比较也呈现同样趋势。这进一步支持了以下直觉：卷积的归纳偏置对小数据集有帮助；而数据量较大时，直接从数据中学习相关模式已经足够，甚至更有利。

总体而言，ImageNet 上的少样本结果（图4）与 VTAB 上的少数据结果（表2），都显示出极少数据迁移的潜力。进一步分析 ViT 的少样本特性，是值得探索的未来方向。

4.4 SCALING STUDY

We perform a controlled scaling study of different models by evaluating transfer performance from JFT-300M. In this setting data size does not bottleneck the models' performances, and we assess performance versus pre-training cost of each model. The model set includes: 7 ResNets, R50x1, R50x2 R101x1, R152x1, R152x2, pre-trained for 7 epochs, plus R152x2 and R200x3 pre-trained for 14 epochs; 6 Vision Transformers, ViT-B/32, B/16, L/32, L/16, pre-trained for 7 epochs, plus L/16 and H/14 pre-trained for 14 epochs; and 5 hybrids, R50+ViT-B/32, B/16, L/32, L/16 pre-trained for 7 epochs, plus R50+ViT-L/16 pre-trained for 14 epochs (for hybrids, the number at the end of the model name stands not for the patch size, but for the total dowsampling ratio in the ResNet backbone).

Figure 5 contains the transfer performance versus total pre-training compute (see Appendix D.5 for details on computational costs). Detailed results per model are provided in Table 6 in the Appendix. A few patterns can be observed. First, Vision Transformers dominate ResNets on the performance/compute trade-off. ViT uses approximately $2 - 4\times$ less compute to attain the same performance (average over 5 datasets). Second, hybrids slightly outperform ViT at small computational budgets, but the difference vanishes for larger models. This result is somewhat surprising, since one might expect convolutional local feature processing to assist ViT at any size. Third, Vision Transformers appear not to saturate within the range tried, motivating future scaling efforts.

4.5 INSPECTING VISION TRANSFORMER

To begin to understand how the Vision Transformer processes image data, we analyze its internal representations. The first layer of the Vision Transformer linearly projects the flattened patches into a lower-dimensional space (Eq. 1). Figure 7 (left) shows the top principal components of the the learned embedding filters. The components resemble plausible basis functions for a low-dimensional representation of the fine structure within each patch.

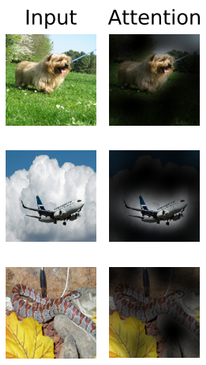


Figure 6: Representative examples of attention from the output token to the input space. See Appendix D.7 for details.

After the projection, a learned position embedding is added to the patch representations. Figure 7 (center) shows that the model learns to encode distance within the image in the similarity of position embeddings, i.e. closer patches tend to have more similar position embeddings. Further, the row-column structure appears; patches in the same row/column have similar embeddings. Finally, a sinusoidal structure is sometimes apparent for larger grids (Appendix D). That the position embeddings learn to represent 2D image topology explains why hand-crafted 2D-aware embedding variants do not yield improvements (Appendix D.4).

Self-attention allows ViT to integrate information across the entire image even in the lowest layers. We investigate to what degree the network makes use of this capability. Specifically, we compute the average distance in image space across which information is integrated, based on the attention weights (Figure 7, right). This "attention distance" is analogous to receptive field size in CNNs. We find that some heads attend to most of the image already in the lowest layers, showing that the ability to integrate information globally is indeed used by the model. Other attention heads have consistently small attention distances in the low layers. This highly localized attention is less pronounced in hybrid models that apply a ResNet before the Transformer (Figure 7, right), suggesting that it may serve a similar function as early convolutional layers in CNNs. Further, the attention distance increases with network depth. Globally, we find that the model attends to image regions that are semantically relevant for classification (Figure 6).

4.6 SELF-SUPERVISION

Transformers show impressive performance on NLP tasks. However, much of their success stems not only from their excellent scalability but also from large scale self-supervised pre-training (Devlin

4.4 规模扩展研究

我们通过评估从 JFT-300M 迁移的性能，对不同模型进行受控的规模扩展研究。在这一设置中，数据量不会成为性能瓶颈，因此可以考察各模型的性能与预训练开销之间的关系。模型包括：7个 ResNet，即预训练7轮的 R50x1、R50x2、R101x1、R152x1、R152x2，以及预训练14轮的 R152x2 和 R200x3；6个视觉 Transformer，即预训练7轮的 ViT-B/32、B/16、L/32、L/16，以及预训练14轮的 L/16 和 H/14；还有5个混合模型，即预训练7轮的 R50+ViT-B/32、B/16、L/32、L/16，以及预训练14轮的 R50+ViT-L/16。混合模型名称末尾的数字表示 ResNet 主干中的总下采样比例，而非图像块大小。

图5展示了迁移性能与预训练总计算量的关系，计算开销详见附录D.5。各模型的详细结果见附录表6。可以观察到几个趋势。第一，在性能与计算量的权衡上，视觉 Transformer 优于 ResNet。达到相同性能时，ViT 的计算量约减少 $2 - 4\times$，这一结果取5个数据集的平均值。第二，在较小计算预算下，混合模型略优于 ViT，但模型更大时，这一差异消失。这有些出乎意料，因为人们可能会认为，卷积的局部特征处理在任何规模下都有助于 ViT。第三，在我们尝试的范围内，视觉 Transformer 的性能似乎尚未饱和，这为继续扩大规模提供了动力。

4.5 观察视觉 Transformer 内部

为初步理解视觉 Transformer 如何处理图像，我们分析了它的内部表征。第一层将展平后的图像块线性投影到更低维空间（式1）。图7左侧显示了学到的嵌入滤波器的主要主成分。这些成分看起来像合理的基函数，可以低维表示每个图像块内部的细致结构。

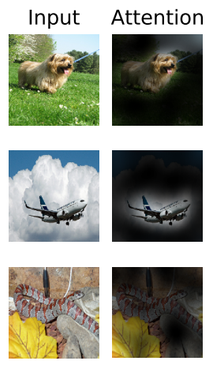


图6：从输出词元到输入空间的注意力代表性示例。详见附录D.7。

投影之后，我们将可学习的位置嵌入加到图像块表征上。图7中间表明，模型通过位置嵌入的相似性编码图像中的距离：距离较近的图像块，其位置嵌入通常更相似。此外，还能观察到行列结构，同一行或同一列的图像块具有相似嵌入。最后，在更大的网格中，有时还能观察到正弦结构（附录D）。位置嵌入能够学会表示图像的二维拓扑，这解释了为什么手工设计、显式考虑二维结构的嵌入变体并未带来改进（附录D.4）。

自注意力使 ViT 即使在最低层，也能整合整张图像的信息。我们研究网络究竟在多大程度上利用了这一能力。具体而言，依据注意力权重，计算图像空间中信息整合的平均距离（图7右）。这一“注意力距离”类似 CNN 的感受野大小。

我们发现，有些注意力头在最低层就已关注图像的大部分区域，说明模型确实利用了全局信息整合能力。其他注意力头在较低层始终保持较短的注意力距离。在 Transformer 前使用 ResNet 的混合模型中，这种高度局部化的注意力不那么明显（图7右），提示它可能发挥了类似 CNN 早期卷积层的作用。此外，注意力距离随网络深度增加而增大。总体而言，模型关注的是在语义上与分类有关的图像区域（图6）。

4.6 自监督

Transformer 在 NLP 任务上表现突出。不过，其成功不仅来自优异的可扩展性，也来自大规模自监督预训练（Devlin

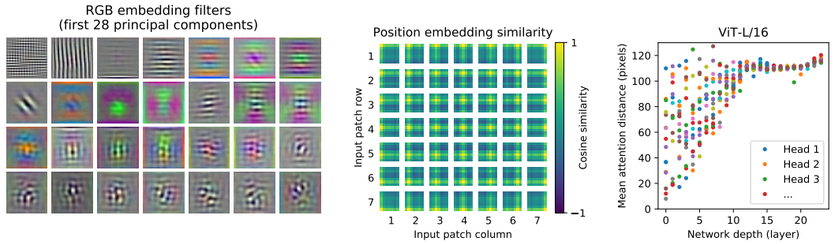


Figure 7: **Left:** Filters of the initial linear embedding of RGB values of ViT-L/32. **Center:** Similarity of position embeddings of ViT-L/32. Tiles show the cosine similarity between the position embedding of the patch with the indicated row and column and the position embeddings of all other patches. **Right:** Size of attended area by head and network depth. Each dot shows the mean attention distance across images for one of 16 heads at one layer. See Appendix D.7 for details.

et al., 2019; Radford et al., 2018). We also perform a preliminary exploration on *masked patch prediction* for self-supervision, mimicking the masked language modeling task used in BERT. With self-supervised pre-training, our smaller ViT-B/16 model achieves 79.9% accuracy on ImageNet, a significant improvement of 2% to training from scratch, but still 4% behind supervised pre-training. Appendix B.1.2 contains further details. We leave exploration of contrastive pre-training (Chen et al., 2020b; He et al., 2020; Bachman et al., 2019; Hénaff et al., 2020) to future work.

5 Conclusion

We have explored the direct application of Transformers to image recognition. Unlike prior works using self-attention in computer vision, we do not introduce image-specific inductive biases into the architecture apart from the initial patch extraction step. Instead, we interpret an image as a sequence of patches and process it by a standard Transformer encoder as used in NLP. This simple, yet scalable, strategy works surprisingly well when coupled with pre-training on large datasets. Thus, Vision Transformer matches or exceeds the state of the art on many image classification datasets, whilst being relatively cheap to pre-train.

While these initial results are encouraging, many challenges remain. One is to apply ViT to other computer vision tasks, such as detection and segmentation. Our results, coupled with those in Carion et al. (2020), indicate the promise of this approach. Another challenge is to continue exploring self-supervised pre-training methods. Our initial experiments show improvement from self-supervised pre-training, but there is still large gap between self-supervised and large-scale supervised pre-training. Finally, further scaling of ViT would likely lead to improved performance.

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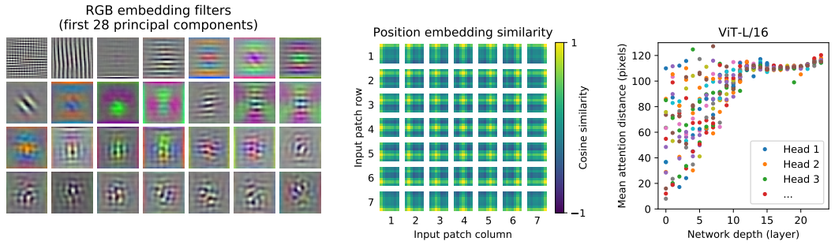


图7：左：ViT-L/32 对 RGB 值进行初始线性嵌入的滤波器。中：ViT-L/32 位置嵌入的相似性。每个小图展示指定行列图像块的位置嵌入，与其他所有图像块位置嵌入之间的余弦相似性。右：不同注意力头和网络深度对应的关注区域大小。每个点表示某层16个注意力头之一，在不同图像上的平均注意力距离。详见附录D.7。

等，2019；Radford 等，2018）。我们还初步探索了用于自监督的掩码图像块预测，模仿 BERT 的掩码语言建模任务。通过自监督预训练，较小的 ViT-B/16 在 ImageNet 上达到79.9%准确率，比从头训练显著提高2个百分点，但仍比监督预训练低4个百分点。更多细节见附录 B.1.2。对比式预训练（Chen 等，2020b；He 等，2020；Bachman 等，2019；H´enaff 等，2020）的探索留待未来工作。

5 结论

我们探索了将 Transformer 直接用于图像识别。与此前在视觉中使用自注意力的工作不同，除了最初的图像块提取步骤，我们没有向架构中加入针对图像的归纳偏置。相反，我们把图像视为图像块序列，并使用 NLP 中的标准 Transformer 编码器处理。结合大型数据集上的预训练，这种简单而可扩展的策略取得了出乎意料的好效果。因此，视觉 Transformer 在多个图像分类数据集上达到或超过最先进的方法，同时预训练成本相对较低。

尽管初步结果令人鼓舞，仍有许多挑战。其一，是将 ViT 应用于检测、分割等其他计算机视觉任务。我们的结果与 Carion 等（2020）的结果共同显示了这一方向的潜力。另一项挑战是继续探索自监督预训练方法。初步实验表明自监督预训练能带来改进，但它与大规模监督预训练之间仍有明显差距。最后，进一步扩大 ViT 的规模，很可能继续提升性能。

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Models	Dataset	Epochs	Base LR	LR decay	Weight decay	Dropout
ViT-B/{16,32}	JFT-300M	7	$8 \cdot 10^{-4}$	linear	0.1	0.0
ViT-L/32	JFT-300M	7	$6 \cdot 10^{-4}$	linear	0.1	0.0
ViT-L/16	JFT-300M	7/14	$4 \cdot 10^{-4}$	linear	0.1	0.0
ViT-H/14	JFT-300M	14	$3 \cdot 10^{-4}$	linear	0.1	0.0
R50x{1,2}	JFT-300M	7	10^{-3}	linear	0.1	0.0
R101x1	JFT-300M	7	$8 \cdot 10^{-4}$	linear	0.1	0.0
R152x{1,2}	JFT-300M	7	$6 \cdot 10^{-4}$	linear	0.1	0.0
R50+ViT-B/{16,32}	JFT-300M	7	$8 \cdot 10^{-4}$	linear	0.1	0.0
R50+ViT-L/32	JFT-300M	7	$2 \cdot 10^{-4}$	linear	0.1	0.0
R50+ViT-L/16	JFT-300M	7/14	$4 \cdot 10^{-4}$	linear	0.1	0.0
ViT-B/{16,32}	ImageNet-21k	90	10^{-3}	linear	0.03	0.1
ViT-L/{16,32}	ImageNet-21k	30/90	10^{-3}	linear	0.03	0.1
ViT-$*$	ImageNet	300	$3 \cdot 10^{-3}$	cosine	0.3	0.1

Table 3: Hyperparameters for training. All models are trained with a batch size of 4096 and learning rate warmup of 10k steps. For ImageNet we found it beneficial to additionally apply gradient clipping at global norm 1. Training resolution is 224.

APPENDIX

A MULTIHEAD SELF-ATTENTION

Standard **qkv** self-attention (SA, Vaswani et al. (2017)) is a popular building block for neural architectures. For each element in an input sequence $\mathbf{z} \in \mathbb{R}^{N \times D}$, we compute a weighted sum over all values $\mathbf{v}$ in the sequence. The attention weights A_{ij} are based on the pairwise similarity between two elements of the sequence and their respective query $\mathbf{q}^i$ and key $\mathbf{k}^j$ representations.

$$[\mathbf{q}, \mathbf{k}, \mathbf{v}] = \mathbf{z}\mathbf{U}_{qkv} \qquad \mathbf{U}_{qkv} \in \mathbb{R}^{D \times 3D_h}, \tag{5}$$

$$A = \mathrm{softmax}\left(\mathbf{q}\mathbf{k}^\top / \sqrt{D_h}\right) \qquad A \in \mathbb{R}^{N \times N}, \tag{6}$$

$$\mathrm{SA}(\mathbf{z}) = A\mathbf{v}\,. \tag{7}$$

Multihead self-attention (MSA) is an extension of SA in which we run k self-attention operations, called "heads", in parallel, and project their concatenated outputs. To keep compute and number of parameters constant when changing k, D_h (Eq. 5) is typically set to D/k.

$$\mathrm{MSA}(\mathbf{z}) = [\mathrm{SA}_1(z); \mathrm{SA}_2(z); \cdots; \mathrm{SA}_k(z)]\,\mathbf{U}_{msa} \qquad \mathbf{U}_{msa} \in \mathbb{R}^{k \cdot D_h \times D} \tag{8}$$

B EXPERIMENT DETAILS

B.1 TRAINING

Table 3 summarizes our training setups for our different models. We found strong regularization to be key when training models from scratch on ImageNet. Dropout, when used, is applied after every dense layer except for the the qkv-projections and directly after adding positional- to patch embeddings. Hybrid models are trained with the exact setup as their ViT counterparts. Finally, all training is done on resolution 224.

B.1.1 FINE-TUNING

We fine-tune all ViT models using SGD with a momentum of 0.9. We run a small grid search over learning rates, see learning rate ranges in Table 4. To do so, we use small sub-splits from the training set (10% for Pets and Flowers, 2% for CIFAR, 1% ImageNet) as development set and train on the remaining data. For final results we train on the entire training set and evaluate on the respective test data. For fine-tuning ResNets and hybrid models we use the exact same setup, with the only exception of ImageNet where we add another value 0.06 to the learning rate sweep. Additionally,

模型	数据集	轮数	基础学习率	学习率衰减	权重衰减	Dropout
ViT-B/{16,32}	JFT-300M	7	$8 \cdot 10^{-4}$	线性	0.1	0.0
ViT-L/32	JFT-300M	7	$6 \cdot 10^{-4}$	线性	0.1	0.0
ViT-L/16	JFT-300M	7/14	$4 \cdot 10^{-4}$	线性	0.1	0.0
ViT-H/14	JFT-300M	14	$3 \cdot 10^{-4}$	线性	0.1	0.0
R50x{1,2}	JFT-300M	7	10^{-3}	线性	0.1	0.0
R101x1	JFT-300M	7	$8 \cdot 10^{-4}$	线性	0.1	0.0
R152x{1,2}	JFT-300M	7	$6 \cdot 10^{-4}$	线性	0.1	0.0
R50+ViT-B/{16,32}	JFT-300M	7	$8 \cdot 10^{-4}$	线性	0.1	0.0
R50+ViT-L/32	JFT-300M	7	$2 \cdot 10^{-4}$	线性	0.1	0.0
R50+ViT-L/16	JFT-300M	7/14	$4 \cdot 10^{-4}$	线性	0.1	0.0
ViT-B/{16,32}	ImageNet-21k	90	10^{-3}	线性	0.03	0.1
ViT-L/{16,32}	ImageNet-21k	30/90	10^{-3}	线性	0.03	0.1
ViT-$*$	ImageNet	300	$3 \cdot 10^{-3}$	余弦	0.3	0.1

表3：训练超参数。所有模型的批量大小均为4096，学习率预热1万步。在 ImageNet 上，我们发现，将梯度全局范数裁剪到1还能带来收益。训练分辨率为224。

附录

A 多头自注意力

标准 qkv 自注意力（SA；Vaswani 等，2017）是神经网络架构中常用的组成模块。对输入序列 $\mathrm{z} \in \mathbb{R}^{N \times D}$ 中的每个元素，我们计算序列中所有值 v 的加权和。注意力权重 A_{ij} 取决于序列中两个元素及其查询 q^i 与键 k^j 表征之间的两两相似性。

$$[\mathbf{q}, \mathbf{k}, \mathbf{v}] = \mathbf{z}\mathbf{U}_{qkv} \qquad \mathbf{U}_{qkv} \in \mathbb{R}^{D \times 3D_h}, \tag{5}$$

$$A = \mathrm{softmax}\left(\mathbf{q}\mathbf{k}^\top / \sqrt{D_h}\right) \qquad A \in \mathbb{R}^{N \times N}, \tag{6}$$

$$\mathrm{SA(z)} = A\mathrm{v}\,. \tag{7}$$

多头自注意力（MSA）是 SA 的扩展：并行执行 k 个称为“头”的自注意力操作，将它们的输出拼接后再投影。为在改变 k 时保持计算量和参数量不变，通常将 D_h（式5）设为 D/k。

$$\mathrm{MSA}(\mathbf{z}) = [\mathrm{SA}_1(z); \mathrm{SA}_2(z); \cdots; \mathrm{SA}_k(z)]\,\mathbf{U}_{msa} \qquad \mathbf{U}_{msa} \in \mathbb{R}^{k \cdot D_h \times D} \tag{8}$$

B 实验细节

B.1 训练

表3汇总了不同模型的训练设置。我们发现，在 ImageNet 上从头训练时，强正则化至关重要。使用 dropout 时，除了 qkv 投影层以外，每个全连接层后都应用 dropout；在将位置嵌入加到图像块嵌入后，也立即应用 dropout。混合模型使用与对应 ViT 完全相同的设置。最后，所有预训练均采用224分辨率。

B.1.1 微调

所有 ViT 模型均使用动量为0.9的 SGD 微调。我们对学习率进行小规模网格搜索，范围见表4。具体而言，从训练集中划出小型开发集：Pets 和 Flowers 为10%，CIFAR 为2%，ImageNet 为1%；使用剩余数据训练。报告最终结果时，使用完整训练集训练，再在对应测试集上评估。ResNet 和混合模型的微调采用完全相同的设置，唯一例外是在 ImageNet 的学习率搜索中额外加入0.06。此外，

Dataset	Steps	Base LR
ImageNet	20 000	{0.003, 0.01, 0.03, 0.06}
CIFAR100	10 000	{0.001, 0.003, 0.01, 0.03}
CIFAR10	10 000	{0.001, 0.003, 0.01, 0.03}
Oxford-IIIT Pets	500	{0.001, 0.003, 0.01, 0.03}
Oxford Flowers-102	500	{0.001, 0.003, 0.01, 0.03}
VTAB (19 tasks)	2 500	0.01

Table 4: Hyperparameters for fine-tuning. All models are fine-tuned with cosine learning rate decay, a batch size of 512, no weight decay, and grad clipping at global norm 1. If not mentioned otherwise, fine-tuning resolution is 384.

for ResNets we also run the setup of Kolesnikov et al. (2020) and select the best results across this run and our sweep. Finally, if not mentioned otherwise, all fine-tuning experiments run at 384 resolution (running fine-tuning at different resolution than training is common practice (Kolesnikov et al., 2020)).

When transferring ViT models to another dataset, we remove the whole head (two linear layers) and replace it by a single, zero-initialized linear layer outputting the number of classes required by the target dataset. We found this to be a little more robust than simply re-initializing the very last layer.

For VTAB we follow the protocol in Kolesnikov et al. (2020), and use the same hyperparameter setting for all tasks. We use a learning rate of 0.01 and train for 2500 steps (Tab. 4). We chose this setting by running a small sweep over two learning rates and two schedules, and selecting the setting with the highest VTAB score on the 200-example validation sets. We follow the pre-processing used in Kolesnikov et al. (2020), except that we do not use task-specific input resolutions. Instead we find that Vision Transformer benefits most from a high resolution (384×384) for all tasks.

B.1.2 Self-supervision

We employ the *masked patch prediction* objective for preliminary self-supervision experiments. To do so we corrupt 50% of patch embeddings by either replacing their embeddings with a learnable `[mask]` embedding (80%), a random other patch embedding (10%) or just keeping them as is (10%). This setup is very similar to the one used for language by Devlin et al. (2019). Finally, we predict the 3-bit, mean color (i.e., 512 colors in total) of every corrupted patch using their respective patch representations.

We trained our self-supervised model for 1M steps (ca. 14 epochs) with batch size 4096 on JFT. We use Adam, with a base learning rate of $2 \cdot 10^{-4}$, warmup of 10k steps and cosine learning rate decay. As prediction targets for pretraining we tried the following settings: 1) predicting only the mean, 3bit color (i.e., 1 prediction of 512 colors), 2) predicting a 4×4 downsized version of the 16×16 patch with 3bit colors in parallel (i.e., 16 predictions of 512 colors), 3) regression on the full patch using L2 (i.e., 256 regressions on the 3 RGB channels). Surprisingly, we found that all worked quite well, though L2 was slightly worse. We report final results only for option 1) because it has shown best few-shot performance. We also experimented with 15% corruption rate as used by Devlin et al. (2019) but results were also slightly worse on our few-shot metrics.

Lastly, we would like to remark that our instantiation of masked patch prediction doesn't require such an enormous amount of pretraining nor a large dataset such as JFT in order to lead to similar performance gains on ImageNet classification. That is, we observed diminishing returns on downstream performance after 100k pretraining steps, and see similar gains when pretraining on ImageNet.

C Additional Results

We report detailed results corresponding to the figures presented in the paper. Table 5 corresponds to Figure 3 from the paper and shows transfer performance of different ViT models pre-trained on datasets of increasing size: ImageNet, ImageNet-21k, and JFT-300M. Table 6 corresponds to

数据集	步数	基础学习率
ImageNet	20 000	{0.003, 0.01, 0.03, 0.06}
CIFAR100	10 000	{0.001, 0.003, 0.01, 0.03}
CIFAR10	10 000	{0.001, 0.003, 0.01, 0.03}
Oxford-IIIT Pets	500	{0.001, 0.003, 0.01, 0.03}
Oxford Flowers-102	500	{0.001, 0.003, 0.01, 0.03}
VTAB（19个任务）	2 500	0.01

表4：微调超参数。所有模型均采用余弦学习率衰减，批量大小512，不使用权重衰减，并将梯度全局范数裁剪到1。除非另有说明，微调分辨率为384。

对于 ResNet，我们还运行 Kolesnikov 等（2020）的设置，并在该运行与我们的参数搜索之间选取最佳结果。最后，除非另有说明，全部微调实验都使用384分辨率。微调采用不同于预训练的分辨率，是常见做法（Kolesnikov 等，2020）。

将 ViT 迁移到其他数据集时，我们移除整个分类头，也就是两个线性层，替换为单个零初始化的线性层，其输出维度等于目标数据集的类别数。我们发现，这比只重新初始化最后一层略稳健。

对于 VTAB，我们遵循 Kolesnikov 等（2020）的协议，并对所有任务采用相同超参数。学习率为0.01，训练2500步（表4）。我们通过对两种学习率和两种训练安排进行小规模搜索，在每个任务包含200个样本的验证集上，选择 VTAB 得分最高的设置。预处理也沿用 Kolesnikov 等（2020），但不使用针对任务的输入分辨率。相反，我们发现，所有任务采用高分辨率（384×384），最有利于视觉 Transformer。

B.1.2 自监督

在初步自监督实验中，我们采用掩码图像块预测目标。具体而言，损坏50%的图像块嵌入：其中80%替换为可学习的 [mask] 嵌入，10%替换为其他随机图像块的嵌入，10%保持不变。这与 Devlin 等（2019）用于语言的设置非常相似。最后，利用每个被损坏图像块的表征，预测该块的3比特平均颜色，共512种颜色。

我们在 JFT 上以4096的批量大小训练自监督模型100万步，约14轮。优化器为 Adam，基础学习率为 $2 \cdot 10^{-4}$，预热1万步，随后采用余弦学习率衰减。我们尝试了三种预训练预测目标：（1）仅预测采用3比特色彩的平均颜色，即在512种颜色中进行1次预测；（2）并行预测 16×16 图像块缩小为 4×4 后的3比特色彩图像，即在512种颜色中进行16次预测；（3）使用 L2 对完整图像块进行回归，即对 3 个 RGB 通道分别进行 256 次回归。令人意外的是，这些方案都表现不错，虽然 L2 略差。我们只报告方案（1）的最终结果，因为它的少样本表现最佳。我们也尝试了 Devlin 等（2019）采用的15%损坏比例，但少样本指标略差。

最后需要指出，我们采用的掩码图像块预测方案，并不需要如此大量的预训练，也不需要像 JFT 这样大的数据集，就能在 ImageNet 分类上获得相近提升。也就是说，预训练超过10万步后，下游性能的收益已开始递减；在 ImageNet 上预训练，也观察到相近收益。

C 补充结果

我们报告了与正文图对应的详细结果。表5对应正文图3，展示不同 ViT 在规模逐步增大的 ImageNet、ImageNet-21k 和 JFT-300M 上预训练后的迁移性能。表6对应

		ViT-B/16	ViT-B/32	ViT-L/16	ViT-L/32	ViT-H/14
ImageNet	CIFAR-10	98.13	97.77	97.86	97.94	-
	CIFAR-100	87.13	86.31	86.35	87.07	-
	ImageNet	77.91	73.38	76.53	71.16	-
	ImageNet ReaL	83.57	79.56	82.19	77.83	-
	Oxford Flowers-102	89.49	85.43	89.66	86.36	-
	Oxford-IIIT-Pets	93.81	92.04	93.64	91.35	-
ImageNet-21k	CIFAR-10	98.95	98.79	99.16	99.13	99.27
	CIFAR-100	91.67	91.97	93.44	93.04	93.82
	ImageNet	83.97	81.28	85.15	80.99	85.13
	ImageNet ReaL	88.35	86.63	88.40	85.65	88.70
	Oxford Flowers-102	99.38	99.11	99.61	99.19	99.51
	Oxford-IIIT-Pets	94.43	93.02	94.73	93.09	94.82
JFT-300M	CIFAR-10	99.00	98.61	99.38	99.19	99.50
	CIFAR-100	91.87	90.49	94.04	92.52	94.55
	ImageNet	84.15	80.73	87.12	84.37	88.04
	ImageNet ReaL	88.85	86.27	89.99	88.28	90.33
	Oxford Flowers-102	99.56	99.27	99.56	99.45	99.68
	Oxford-IIIT-Pets	95.80	93.40	97.11	95.83	97.56

Table 5: Top1 accuracy (in %) of Vision Transformer on various datasets when pre-trained on ImageNet, ImageNet-21k or JFT300M. These values correspond to Figure 3 in the main text. Models are fine-tuned at 384 resolution. Note that the ImageNet results are computed without additional techniques (Polyak averaging and 512 resolution images) used to achieve results in Table 2.

name	Epochs	ImageNet	ImageNet ReaL	CIFAR-10	CIFAR-100	Pets	Flowers	exaFLOPs
ViT-B/32	7	80.73	86.27	98.61	90.49	93.40	99.27	55
ViT-B/16	7	84.15	88.85	99.00	91.87	95.80	99.56	224
ViT-L/32	7	84.37	88.28	99.19	92.52	95.83	99.45	196
ViT-L/16	7	86.30	89.43	99.38	93.46	96.81	99.66	783
ViT-L/16	14	87.12	89.99	99.38	94.04	97.11	99.56	1567
ViT-H/14	14	88.08	90.36	99.50	94.71	97.11	99.71	4262
ResNet50x1	7	77.54	84.56	97.67	86.07	91.11	94.26	50
ResNet50x2	7	82.12	87.94	98.29	89.20	93.43	97.02	199
ResNet101x1	7	80.67	87.07	98.48	89.17	94.08	95.95	96
ResNet152x1	7	81.88	87.96	98.82	90.22	94.17	96.94	141
ResNet152x2	7	84.97	89.69	99.06	92.05	95.37	98.62	563
ResNet152x2	14	85.56	89.89	99.24	91.92	95.75	98.75	1126
ResNet200x3	14	87.22	90.15	99.34	93.53	96.32	99.04	3306
R50x1+ViT-B/32	7	84.90	89.15	99.01	92.24	95.75	99.46	106
R50x1+ViT-B/16	7	85.58	89.65	99.14	92.63	96.65	99.40	274
R50x1+ViT-L/32	7	85.68	89.04	99.24	92.93	96.97	99.43	246
R50x1+ViT-L/16	7	86.60	89.72	99.18	93.64	97.03	99.40	859
R50x1+ViT-L/16	14	87.12	89.76	99.31	93.89	97.36	99.11	1668

Table 6: Detailed results of model scaling experiments. These correspond to Figure 5 in the main paper. We show transfer accuracy on several datasets, as well as the pre-training compute (in exaFLOPs).

Figure 5 from the paper and shows the transfer performance of ViT, ResNet, and hybrid models of varying size, as well as the estimated computational cost of their pre-training.

D ADDITIONAL ANALYSES

D.1 SGD VS. ADAM FOR RESNETS

ResNets are typically trained with SGD and our use of Adam as optimizer is quite unconventional. Here we show the experiments that motivated this choice. Namely, we compare the fine-tuning

		ViT-B/16	ViT-B/32	ViT-L/16	ViT-L/32	ViT-H/14
ImageNet	CIFAR-10	98.13	97.77	97.86	97.94	-
	CIFAR-100	87.13	86.31	86.35	87.07	-
	ImageNet	77.91	73.38	76.53	71.16	-
	ImageNet ReaL	83.57	79.56	82.19	77.83	-
	Oxford Flowers-102	89.49	85.43	89.66	86.36	-
	Oxford-IIIT-Pets	93.81	92.04	93.64	91.35	-
ImageNet-21k	CIFAR-10	98.95	98.79	99.16	99.13	99.27
	CIFAR-100	91.67	91.97	93.44	93.04	93.82
	ImageNet	83.97	81.28	85.15	80.99	85.13
	ImageNet ReaL	88.35	86.63	88.40	85.65	88.70
	Oxford Flowers-102	99.38	99.11	99.61	99.19	99.51
	Oxford-IIIT-Pets	94.43	93.02	94.73	93.09	94.82
JFT-300M	CIFAR-10	99.00	98.61	99.38	99.19	99.50
	CIFAR-100	91.87	90.49	94.04	92.52	94.55
	ImageNet	84.15	80.73	87.12	84.37	88.04
	ImageNet ReaL	88.85	86.27	89.99	88.28	90.33
	Oxford Flowers-102	99.56	99.27	99.56	99.45	99.68
	Oxford-IIIT-Pets	95.80	93.40	97.11	95.83	97.56

表5：视觉 Transformer 在 ImageNet、ImageNet-21k 或 JFT-300M 上预训练后，在不同数据集上的 top-1 准确率（%）。这些数值对应正文图3。模型以384分辨率微调。请注意，这里的 ImageNet 结果没有采用表2所用的额外技术，即 Polyak 平均和512分辨率图像。

name	轮数	ImageNet	ImageNet ReaL	CIFAR-10	CIFAR-100	Pets	Flowers	exaFLOPs
ViT-B/32	7	80.73	86.27	98.61	90.49	93.40	99.27	55
ViT-B/16	7	84.15	88.85	99.00	91.87	95.80	99.56	224
ViT-L/32	7	84.37	88.28	99.19	92.52	95.83	99.45	196
ViT-L/16	7	86.30	89.43	99.38	93.46	96.81	99.66	783
ViT-L/16	14	87.12	89.99	99.38	94.04	97.11	99.56	1567
ViT-H/14	14	88.08	90.36	99.50	94.71	97.11	99.71	4262
ResNet50x1	7	77.54	84.56	97.67	86.07	91.11	94.26	50
ResNet50x2	7	82.12	87.94	98.29	89.20	93.43	97.02	199
ResNet101x1	7	80.67	87.07	98.48	89.17	94.08	95.95	96
ResNet152x1	7	81.88	87.96	98.82	90.22	94.17	96.94	141
ResNet152x2	7	84.97	89.69	99.06	92.05	95.37	98.62	563
ResNet152x2	14	85.56	89.89	99.24	91.92	95.75	98.75	1126
ResNet200x3	14	87.22	90.15	99.34	93.53	96.32	99.04	3306
R50x1+ViT-B/32	7	84.90	89.15	99.01	92.24	95.75	99.46	106
R50x1+ViT-B/16	7	85.58	89.65	99.14	92.63	96.65	99.40	274
R50x1+ViT-L/32	7	85.68	89.04	99.24	92.93	96.97	99.43	246
R50x1+ViT-L/16	7	86.60	89.72	99.18	93.64	97.03	99.40	859
R50x1+ViT-L/16	14	87.12	89.76	99.31	93.89	97.36	99.11	1668

表6：模型规模扩展实验的详细结果，对应正文图5。我们展示了多个数据集上的迁移准确率，以及预训练计算量，单位为 exaFLOPs。

正文图5，展示不同大小 ViT、ResNet 和混合模型的迁移性能，以及估计的预训练计算开销。

D 补充分析

D.1 ResNet 中的 SGD 与 Adam

ResNet 通常使用 SGD 训练，而我们选择 Adam 的做法很不常规。这里报告促使我们做出这一选择的实验。具体而言，我们比较微调

Dataset	ResNet50 Adam	ResNet50 SGD	ResNet152x2 Adam	ResNet152x2 SGD
ImageNet	77.54	78.24	84.97	84.37
CIFAR10	97.67	97.46	99.06	99.07
CIFAR100	86.07	85.17	92.05	91.06
Oxford-IIIT Pets	91.11	91.00	95.37	94.79
Oxford Flowers-102	94.26	92.06	98.62	99.32
Average	89.33	88.79	94.01	93.72

Table 7: Fine-tuning ResNet models pre-trained with Adam and SGD.

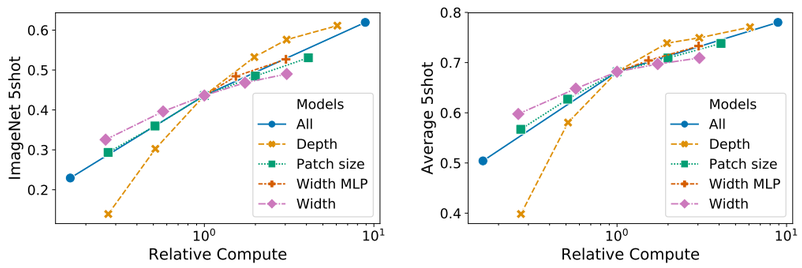


Figure 8: Scaling different model dimensions of the Vision Transformer.

performance of two ResNets – 50x1 and 152x2 – pre-trained on JFT with SGD and Adam. For SGD, we use the hyperparameters recommended by Kolesnikov et al. (2020). Results are presented in Table 7. Adam pre-training outperforms SGD pre-training on most datasets and on average. This justifies the choice of Adam as the optimizer used to pre-train ResNets on JFT. Note that the absolute numbers are lower than those reported by Kolesnikov et al. (2020), since we pre-train only for 7 epochs, not 30.

D.2 Transformer shape

We ran ablations on scaling different dimensions of the Transformer architecture to find out which are best suited for scaling to very large models. Figure 8 shows 5-shot performance on ImageNet for different configurations. All configurations are based on a ViT model with 8 layers, $D = 1024$, $D_{MLP} = 2048$ and a patch size of 32, the intersection of all lines. We can see that scaling the depth results in the biggest improvements which are clearly visible up until 64 layers. However, diminishing returns are already visible after 16 layers. Interestingly, scaling the width of the network seems to result in the smallest changes. Decreasing the patch size and thus increasing the effective sequence length shows surprisingly robust improvements without introducing parameters. These findings suggest that compute might be a better predictor of performance than the number of parameters, and that scaling should emphasize depth over width if any. Overall, we find that scaling all dimensions proportionally results in robust improvements.

D.3 Head Type and `class` token

In order to stay as close as possible to the original Transformer model, we made use of an additional `[class]` token, which is taken as image representation. The output of this token is then transformed into a class prediction via a small multi-layer perceptron (MLP) with tanh as non-linearity in the single hidden layer.

This design is inherited from the Transformer model for text, and we use it throughout the main paper. An initial attempt at using only image-patch embeddings, globally average-pooling (GAP) them, followed by a linear classifier—just like ResNet's final feature map—performed very poorly. However, we found that this is neither due to the extra token, nor to the GAP operation. Instead,

数据集	ResNet50 Adam	ResNet50 SGD	ResNet152x2 Adam	ResNet152x2 SGD
ImageNet	77.54	78.24	84.97	84.37
CIFAR10	97.67	97.46	99.06	99.07
CIFAR100	86.07	85.17	92.05	91.06
Oxford-IIIT Pets	91.11	91.00	95.37	94.79
Oxford Flowers-102	94.26	92.06	98.62	99.32
平均	89.33	88.79	94.01	93.72

表7：对分别使用 Adam 与 SGD 预训练的 ResNet 进行微调。

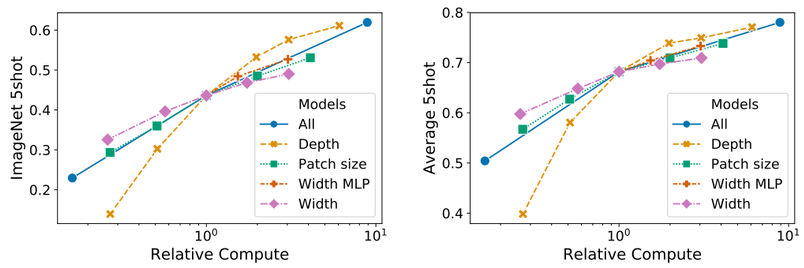


图8：视觉 Transformer 不同模型维度的扩展。

两个 ResNet——50x1 和152x2——分别用 SGD 和 Adam 在 JFT 上预训练后的表现。SGD 使用 Kolesnikov 等（2020）推荐的超参数。结果见表7。在大多数数据集上及平均表现上，Adam 预训练优于 SGD，这支持了我们选择 Adam 在 JFT 上预训练 ResNet。请注意，这里的绝对结果低于 Kolesnikov 等（2020），因为我们只预训练7轮，而非30轮。

D.2 Transformer 的结构尺寸

我们对 Transformer 架构不同维度的扩展进行消融，以确定哪些维度最适合扩展到超大型模型。图8展示了不同配置在 ImageNet 上的5样本性能。所有配置都以一个8层、$D = 1024$、$D_{MLP} = 2048$ 、图像块大小为32的 ViT 为基准，即所有曲线的交点。可以看到，增加深度带来的提升最大，直至64层仍清晰可见；但16层之后已出现收益递减。有意思的是，增加网络宽度带来的变化似乎最小。减小图像块，从而增加有效序列长度，即使不增加参数，也带来了出乎意料的稳定提升。这些发现表明，计算量可能比参数量更能预测性能；若必须有所侧重，规模扩展应优先增加深度，而非宽度。总体而言，我们发现按比例扩展所有维度，可以带来稳定提升。

D.3 分类头类型与分类词元

为尽可能贴近原始 Transformer，我们使用额外的 [class] 词元，并将其作为图像表征。该词元的输出，再通过一个小型多层感知机（MLP）得到类别预测；这个 MLP 只有一个隐藏层，采用 tanh 非线性。

这一设计继承自用于文本的 Transformer，我们在正文所有实验中都使用它。起初，我们尝试只使用图像块嵌入，对它们进行全局平均池化（GAP），再接线性分类器，就像处理 ResNet 的最终特征图一样；结果很差。不过，我们发现问题既不在额外词元，也不在 GAP 操作本身。相反，

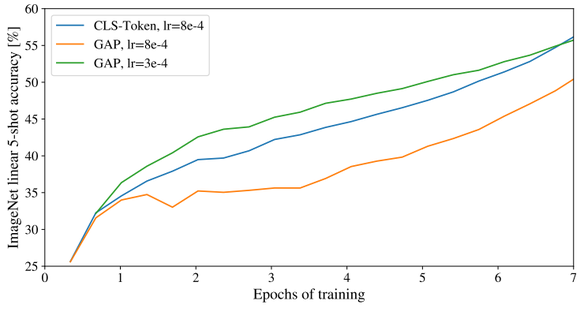


Figure 9: Comparison of class-token and global average pooling classifiers. Both work similarly well, but require different learning-rates.

Pos. Emb.	Default/Stem	Every Layer	Every Layer-Shared
No Pos. Emb.	0.61382	N/A	N/A
1-D Pos. Emb.	0.64206	0.63964	0.64292
2-D Pos. Emb.	0.64001	0.64046	0.64022
Rel. Pos. Emb.	0.64032	N/A	N/A

Table 8: Results of the ablation study on positional embeddings with ViT-B/16 model evaluated on ImageNet 5-shot linear.

the difference in performance is fully explained by the requirement for a different learning-rate, see Figure 9.

D.4 Positional Embedding

We ran ablations on different ways of encoding spatial information using positional embedding. We tried the following cases:

- Providing no positional information: Considering the inputs as a *bag of patches*.
- 1-dimensional positional embedding: Considering the inputs as a sequence of patches in the raster order (default across all other experiments in this paper).
- 2-dimensional positional embedding: Considering the inputs as a grid of patches in two dimensions. In this case, two sets of embeddings are learned, each for one of the axes, X-embedding, and Y-embedding, each with size $D/2$. Then, based on the coordinate on the path in the input, we concatenate the X and Y embedding to get the final positional embedding for that patch.
- Relative positional embeddings: Considering the relative distance between patches to encode the spatial information as instead of their absolute position. To do so, we use 1-dimensional Relative Attention, in which we define the relative distance all possible pairs of patches. Thus, for every given pair (one as query, and the other as key/value in the attention mechanism), we have an offset $p_q - p_k$, where each offset is associated with an embedding. Then, we simply run extra attention, where we use the original query (the content of query), but use relative positional embeddings as keys. We then use the logits from the relative attention as a bias term and add it to the logits of the main attention (content-based attention) before applying the softmax.

In addition to different ways of encoding spatial information, we also tried different ways of incorporating this information in our model. For the 1-dimensional and 2-dimensional positional embeddings, we tried three different cases: (1) add positional embeddings to the inputs right after

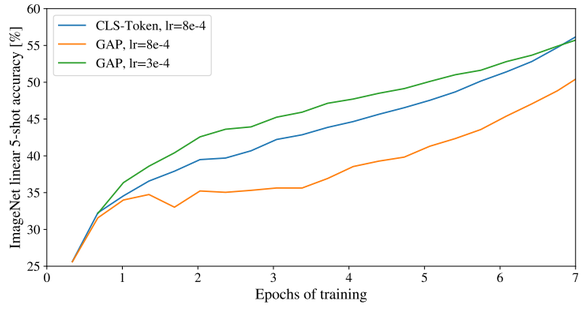


图9：分类词元与全局平均池化分类器的比较。两者表现相近，但需要不同的学习率。

位置嵌入	默认/输入端	每一层	各层共享
无位置嵌入	0.61382	N/A	N/A
一维位置嵌入	0.64206	0.63964	0.64292
二维位置嵌入	0.64001	0.64046	0.64022
相对位置嵌入	0.64032	N/A	N/A

表8：ViT-B/16 位置嵌入消融实验结果，采用 ImageNet 5样本线性评估。

性能差异完全可以由所需学习率不同解释，见图9。

D.4 位置嵌入

我们对使用位置嵌入编码空间信息的不同方式进行了消融，尝试以下设置：

- 不提供位置信息：将输入视为无序的图像块集合。
- 1维位置嵌入：将输入视为按光栅顺序排列的图像块序列。这是本文其他全部实验的默认设置。
- 2维位置嵌入：将输入视为二维图像块网格。此时学习两组嵌入，分别对应两个坐标轴，即 X 嵌入和 Y 嵌入，每组大小为 $D/2$。根据输入中图像块的坐标，拼接对应的 X 和 Y 嵌入，得到该图像块的最终位置嵌入。

- 相对位置嵌入：根据图像块之间的相对距离，而非绝对位置，编码空间信息。为此，我们使用一维相对注意力，并为所有可能的图像块对定义相对距离。于是，对任意一对图像块，一个作为查询，另一个作为注意力机制中的键和值，都存在一个偏移量 $p_q - p_k$，每种偏移量对应一个嵌入。随后，我们额外执行一次注意力计算：查询仍使用原始查询内容，键则使用相对位置嵌入。最后，将相对注意力得到的 logits 作为偏置，加到主要的、基于内容的注意力 logits 上，再应用 softmax。

除了不同空间信息编码方式，我们还尝试以不同方式将这些信息加入模型。对于一维和二维位置嵌入，分别尝试三种设置：（1）在

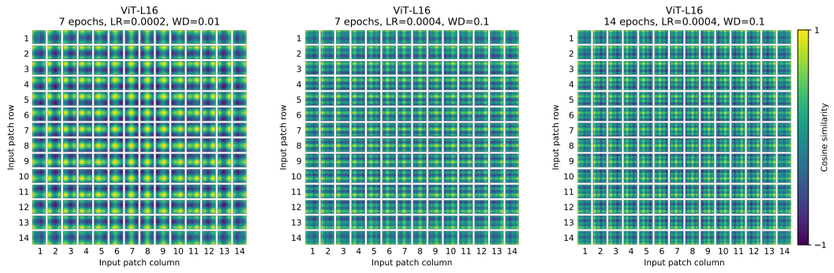


Figure 10: Position embeddings of models trained with different hyperparameters.

the stem of them model and before feeding the inputs to the Transformer encoder (default across all other experiments in this paper); (2) learn and add positional embeddings to the inputs at the beginning of each layer; (3) add a learned positional embeddings to the inputs at the beginning of each layer (shared between layers).

Table 8 summarizes the results from this ablation study on a ViT-B/16 model. As we can see, while there is a large gap between the performances of the model with no positional embedding and models with positional embedding, there is little to no difference between different ways of encoding positional information. We speculate that since our Transformer encoder operates on patch-level inputs, as opposed to pixel-level, the differences in how to encode spatial information is less important. More precisely, in patch-level inputs, the spatial dimensions are much smaller than the original pixel-level inputs, e.g., 14×14 as opposed to 224×224, and learning to represent the spatial relations in this resolution is equally easy for these different positional encoding strategies. Even so, the specific pattern of position embedding similarity learned by the network depends on the training hyperparameters (Figure 10).

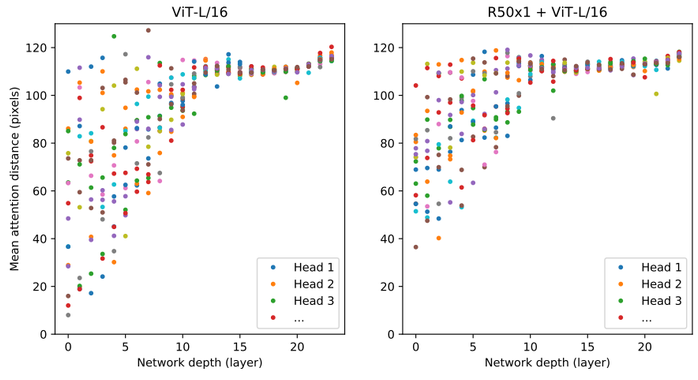


Figure 11: Size of attended area by head and network depth. Attention distance was computed for 128 example images by averaging the distance between the query pixel and all other pixels, weighted by the attention weight. Each dot shows the mean attention distance across images for one of 16 heads at one layer. Image width is 224 pixels.

D.5 Empirical Computational Costs

We are also interested in real-world speed of the architectures on our hardware, which is not always well predicted by theoretical FLOPs due to details like lane widths and cache sizes. For this purpose,

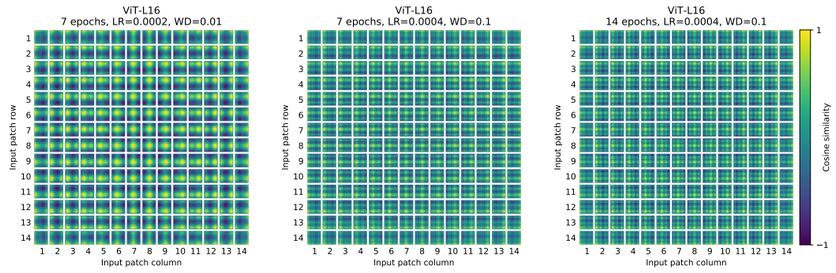


图10：不同超参数训练所得模型的位置嵌入。

模型的输入端，即送入 Transformer 编码器之前添加位置嵌入，这是本文其他实验的默认方式；（2）在每一层开始处，为该层输入学习并添加独立的位置嵌入；（3）在每一层开始处添加学习到的位置嵌入，但各层共享同一组嵌入。

表8汇总了 ViT-B/16 上的这项消融实验。可以看到，不使用位置嵌入的模型与使用位置嵌入的模型之间存在明显性能差距，但不同位置编码方式之间几乎没有差别。我们推测，这是因为 Transformer 编码器处理的是图像块级输入，而非像素级输入，因此空间信息的具体编码方式没有那么重要。更准确地说，图像块级输入的空间尺寸远小于原始像素级输入，例如 14×14 而非 224×224。在这一分辨率下，这些不同的位置编码策略都能同样容易地学会表示空间关系。即便如此，网络学到的位置嵌入相似性模式，仍取决于训练超参数（图10）。

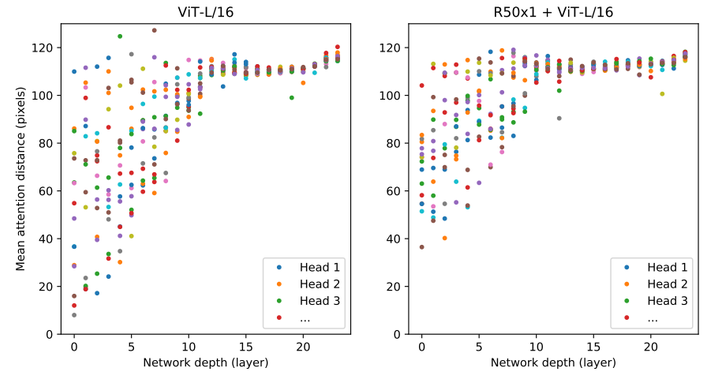


图11：不同注意力头及网络深度对应的关注区域大小。我们对128张样例图像计算注意力距离：求查询像素到其他所有像素的距离，再按注意力权重取加权平均。每个点表示某层16个注意力头之一在各图像上的平均距离。图像宽度为224像素。

D.5 实测计算开销

我们还关心这些架构在所用硬件上的实际速度。由于计算单元宽度、缓存大小等细节，理论 FLOPs 并不总能准确预测实际速度。为此，

we perform timing of inference speed for the main models of interest, on a TPUv3 accelerator; the difference between inference and backprop speed is a constant model-independent factor.

Figure 12 (left) shows how many images one core can handle per second, across various input sizes. Every single point refers to the peak performance measured across a wide range of batch-sizes. As can be seen, the theoretical bi-quadratic scaling of ViT with image size only barely starts happening for the largest models at the largest resolutions.

Another quantity of interest is the largest batch-size each model can fit onto a core, larger being better for scaling to large datasets. Figure 12 (right) shows this quantity for the same set of models. This shows that large ViT models have a clear advantage in terms of memory-efficiency over ResNet models.

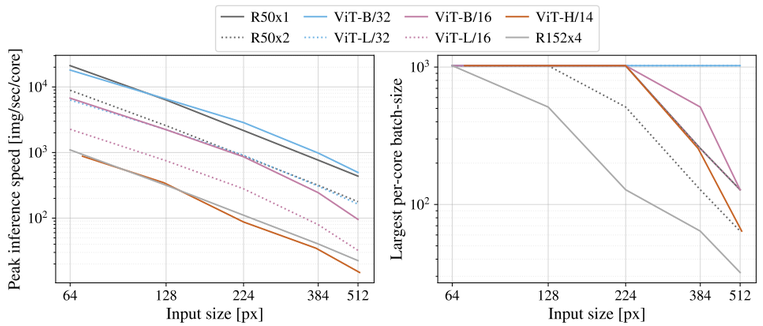


Figure 12: **Left:** Real wall-clock timings of various architectures across input sizes. ViT models have speed comparable to similar ResNets. **Right**: Largest per-core batch-size fitting on device with various architectures across input sizes. ViT models are clearly more memory-efficient.

D.6 Axial Attention

Axial Attention (Huang et al., 2020; Ho et al., 2019) is a simple, yet effective technique to run self-attention on large inputs that are organized as multidimensional tensors. The general idea of axial attention is to perform multiple attention operations, each along a single axis of the input tensor, instead of applying 1-dimensional attention to the flattened version of the input. In axial attention, each attention mixes information along a particular axis, while keeping information along the other axes independent. Along this line, Wang et al. (2020b) proposed the AxialResNet model in which all the convolutions with kernel size 3×3 in a ResNet50 are replaced by axial self-attention, i.e. a row and column attention, augmented by relative positional encoding. We have implemented AxialResNet as a baseline model.[3].

Moreover, we have modified ViT to process inputs in the 2-dimensional shape, instead of a 1-dimensional sequence of patches, and incorporate Axial Transformer blocks, in which instead of a self-attention followed by an MLP, we have a a row-self-attention plus an MLP followed by a column-self-attention plus an MLP.

Figure 13, present the performance of Axial ResNet, Axial-ViT-B/32 and Axial-ViT-B/16 on ImageNet 5shot linear, when pretrained on JFT dataset, verses the pretraining compute, both in terms of number of FLOPs and inference time (example per seconds). As we can see, both Axial-ViT-B/32 and Axial-ViT-B/16 do better than their ViT-B counterpart in terms of performance, but it comes at

[3]Our implementation is based on the open-sourced PyTorch implementation in `https://github.com/csrhddlam/axial-deeplab`. In our experiments, we reproduced the scores reported in (Wang et al., 2020b) in terms of accuracy, however, our implementation, similar to the open-source implementation, is very slow on TPUs. Therefore, we were not able to use it for extensive large-scale experiments. These may be unlocked by a carefully optimized implementation.

我们在 TPUv3 加速器上测量所关注主要模型的推理速度。推理速度与反向传播速度之间，只相差一个与模型无关的常数因子。

图12左侧展示了不同输入尺寸下，每个核心每秒可处理的图像数。每个点表示在广泛批量大小范围内测得的峰值性能。可以看到，ViT 理论上随图像边长呈四次增长的计算开销，只有在最大模型、最高分辨率处，才刚开始显现。

另一个值得关注的量，是每个模型在单个核心上能够容纳的最大批量大小。批量越大，越有利于扩展到大型数据集。图12右侧展示了同一组模型的这一指标。可以看到，大型 ViT 在内存效率上明显优于 ResNet。

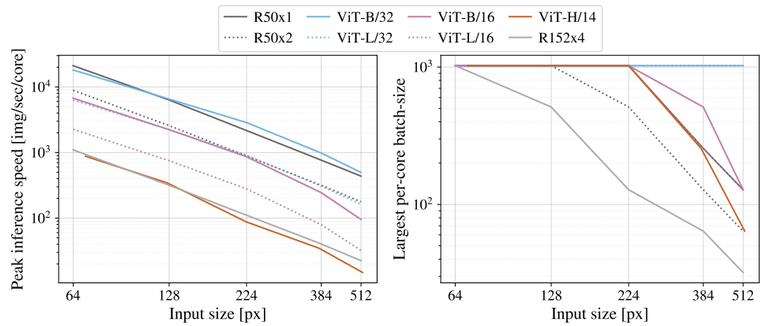


图12：左：不同输入尺寸下，各架构的实际运行速度。ViT 的速度与相近的 ResNet 相当。右：不同输入尺寸下，各架构在单个核心上能容纳的最大批量大小。ViT 的内存效率明显更高。

D.6 轴向注意力

轴向注意力（Huang 等，2020；Ho 等，2019）是一种简单而有效的方法，可对以多维张量组织的大型输入计算自注意力。其基本思路是沿输入张量的各个坐标轴分别执行多次注意力操作，而非将输入展平后计算一维注意力。每次轴向注意力只混合某一轴上的信息，其他轴上的信息仍保持独立。沿着这一思路，Wang 等（2020b）提出了 AxialResNet：将 ResNet50 中所有核大小为 3×3 的卷积替换为轴向自注意力，即结合相对位置编码的行注意力和列注意力。我们实现了 AxialResNet，并将其作为基线模型。[3]。

此外，我们修改 ViT，使其处理二维形状的输入，而非一维图像块序列，并引入轴向 Transformer 块。它不再采用一次自注意力加一个 MLP，而是先进行行自注意力与 MLP，再进行列自注意力与 MLP。

图13展示了 AxialResNet、Axial-ViT-B/32 和 Axial-ViT-B/16 在 JFT 上预训练后，ImageNet 5样本线性评估性能与预训练计算开销之间的关系。计算开销以 FLOPs 及推理吞吐量（每秒样本数）衡量。可以看到，Axial-ViT-B/32 和 Axial-ViT-B/16 都比相应的 ViT-B 表现更好，但代价是

[3]我们的实现以开源 PyTorch 实现 https://github.com/csrhddlam/axial-deeplab 为基础。实验中，我们复现了 Wang 等（2020b）报告的准确率；但与该开源实现相似，我们的版本在 TPU 上非常慢，因此无法用它开展广泛的大规模实验。经过精细优化的实现，可能使这类实验成为可行。

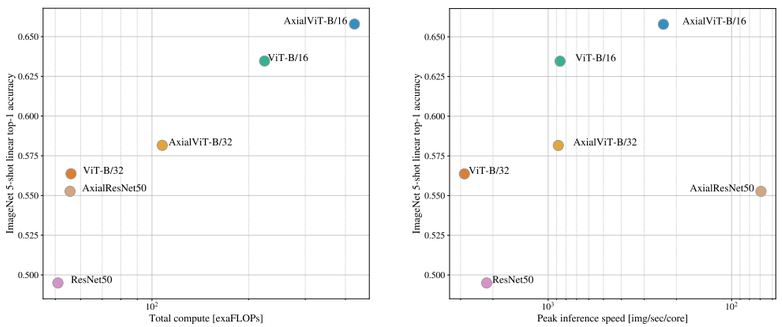


Figure 13: Performance of Axial-Attention based models, in terms of top-1 accuracy on ImageNet 5-shot linear, versus their speed in terms of number of FLOPs (**left**) and inference time (**left**).

the cost of more compute. This is because in Axial-ViT models, each Transformer block with global self-attention is replaced by two Axial Transformer blocks, one with row and one with column self-attention and although the sequence length that self-attention operates on is smaller in axial case, there is a extra MLP per Axial-ViT block. For the AxialResNet, although it looks reasonable in terms of accuracy/compute trade-off (Figure 13, left), the naive implementation is extremely slow on TPUs (Figure 13, right).

D.7 Attention Distance

To understand how ViT uses self-attention to integrate information across the image, we analyzed the average distance spanned by attention weights at different layers (Figure 11). This "attention distance" is analogous to receptive field size in CNNs. Average attention distance is highly variable across heads in lower layers, with some heads attending to much of the image, while others attend to small regions at or near the query location. As depth increases, attention distance increases for all heads. In the second half of the network, most heads attend widely across tokens.

D.8 Attention Maps

To compute maps of the attention from the output token to the input space (Figures 6 and 14), we used Attention Rollout (Abnar & Zuidema, 2020). Briefly, we averaged attention weights of ViT-L/16 across all heads and then recursively multiplied the weight matrices of all layers. This accounts for the mixing of attention across tokens through all layers.

D.9 ObjectNet Results

We also evaluate our flagship ViT-H/14 model on the ObjectNet benchmark following the evaluation setup in Kolesnikov et al. (2020), resulting in 82.1% top-5 accuracy and 61.7% top-1 accuracy.

D.10 VTAB Breakdown

Table 9 shows the scores attained on each of the VTAB-1k tasks.

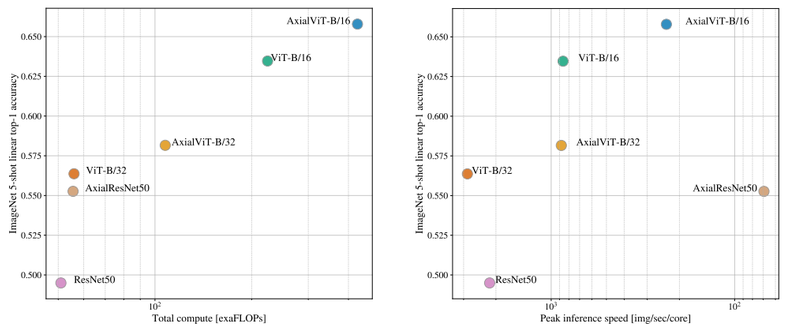


图13：基于轴向注意力的模型，在 ImageNet 5样本线性评估中的 top-1 准确率，与速度之间的关系。速度分别按 FLOPs（左）及推理时间（原文标注为左）衡量。

代价是更多计算。这是因为 Axial-ViT 将每个采用全局自注意力的 Transformer 块，替换为两个轴向 Transformer 块，分别采用行注意力和列注意力。虽然轴向注意力处理的序列更短，但每个 Axial-ViT 块都多了一个 MLP。对于 AxialResNet，虽然其准确率与计算量的权衡看起来合理（图13左），但直接实现版本在 TPU 上极其缓慢（图13右）。

D.7 注意力距离

为理解 ViT 如何通过自注意力整合图像信息，我们分析了不同层中注意力权重覆盖的平均距离（图11）。这种“注意力距离”类似 CNN 的感受野大小。较低层不同注意力头的平均距离差别很大：有些关注图像的大部分区域，有些只关注查询位置附近的小区域。随着深度增加，所有注意力头的距离都增大。在网络后半部分，大多数注意力头广泛关注不同词元。

D.8 注意力图

为计算从输出词元到输入空间的注意力图（图6、14），我们使用 Attention Rollout（Abnar 与 Zuidema，2020）。简而言之，先对 ViT-L/16 所有注意力头的权重取平均，再递归地将各层权重矩阵相乘，以考虑全部层中不同词元之间的注意力混合。

D.9 ObjectNet 结果

我们还按照 Kolesnikov 等（2020）的评估设置，在 ObjectNet 上评估旗舰模型 ViT-H/14，得到 82.1%的 top-5 准确率和61.7%的 top-1 准确率。

D.10 VTAB 性能分解

表9展示各个 VTAB-1k 任务上的得分。

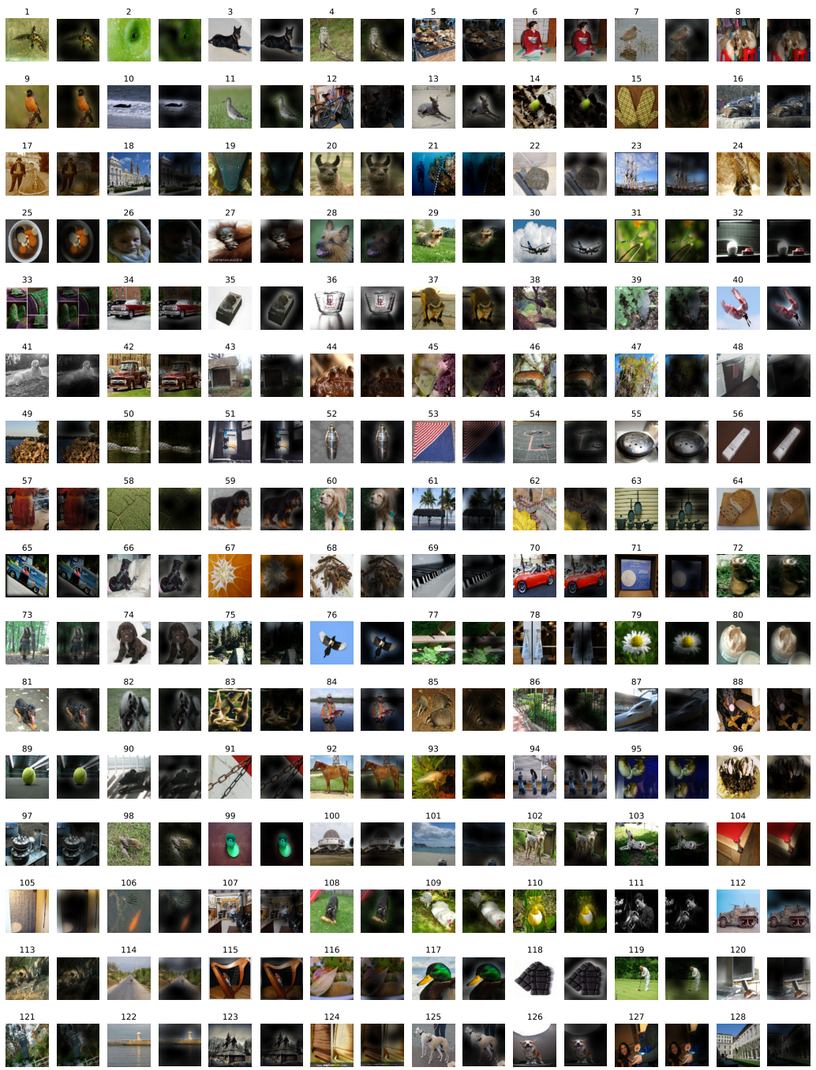


Figure 14: Further example attention maps as in Figure 6 (random selection).

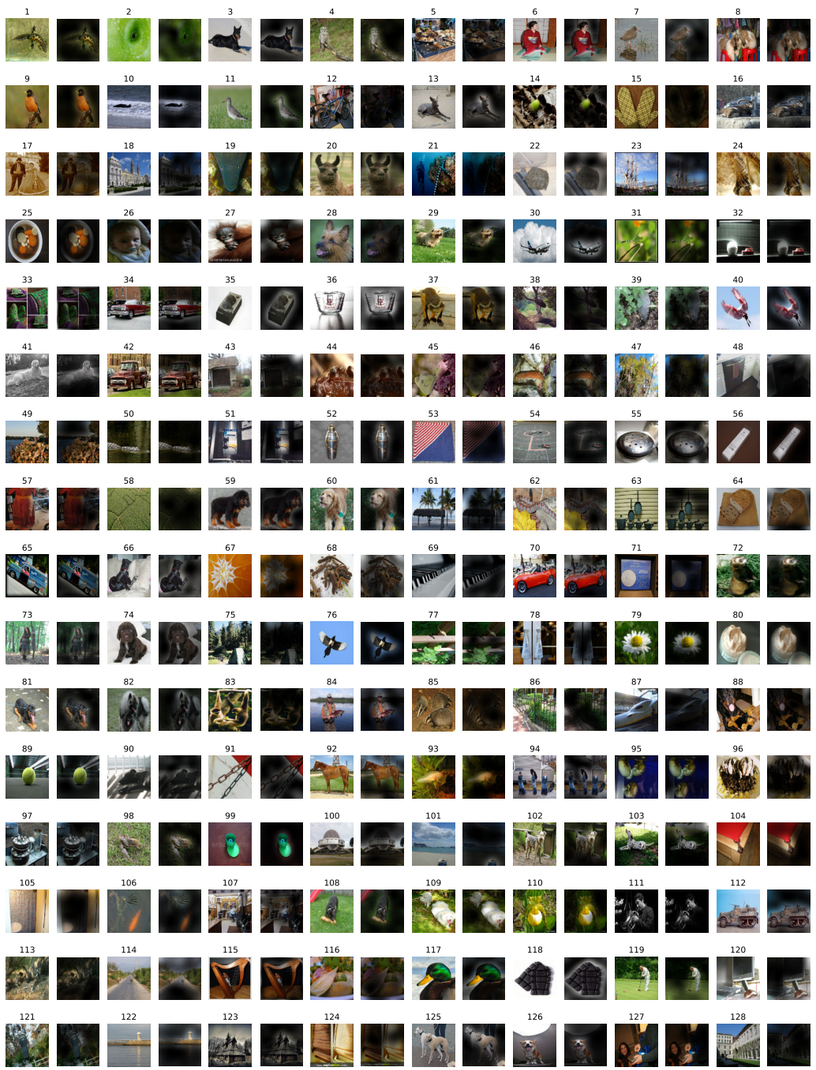


图14：与图6相同形式的更多注意力图示例，随机选取。

Table 9: Breakdown of VTAB-1k performance across tasks.

	Caltech101	CIFAR-100	DTD	Flowers102	Pets	Sun397	SVHN	Camelyon	EuroSAT	Resisc45	Retinopathy	Clevr-Count	Clevr-Dist	DMLab	dSpr-Loc	dSpr-Ori	KITTI-Dist	sNORB-Azim	sNORB-Elev	Mean
ViT-H/14 (JFT)	95.3	85.5	75.2	99.7	97.2	65.0	88.9	83.3	96.7	91.4	76.6	91.7	63.8	53.1	79.4	63.3	84.5	33.2	51.2	77.6
ViT-L/16 (JFT)	95.4	81.9	74.3	99.7	96.7	63.5	87.4	83.6	96.5	89.7	77.1	86.4	63.1	49.7	74.5	60.5	82.2	36.2	51.1	76.3
ViT-L/16 (I21k)	90.8	84.1	74.1	99.3	92.7	61.0	80.9	82.5	95.6	85.2	75.3	70.3	56.1	41.9	74.7	64.9	79.9	30.5	41.7	72.7

表9：各任务上的 VTAB-1k 性能分解。

	Caltech101	CIFAR-100	DTD	Flowers102	Pets	Sun397	SVHN	Camelyon	EuroSAT	Resisc45	Retinopathy	Clevr-Count	Clevr-Dist	DMLab	dSpr-Loc	dSpr-Ori	KITTI-Dist	sNORB-Azim	sNORB-Elev	Mean
ViT-H/14 (JFT)	95.3	85.5	75.2	99.7	97.2	65.0	88.9	83.3	96.7	91.4	76.6	91.7	63.8	53.1	79.4	63.3	84.5	33.2	51.2	77.6
ViT-L/16 (JFT)	95.4	81.9	74.3	99.7	96.7	63.5	87.4	83.6	96.5	89.7	77.1	86.4	63.1	49.7	74.5	60.5	82.2	36.2	51.1	76.3
ViT-L/16(I21k)	90.8	84.1	74.1	99.3	92.7	61.0	80.9	82.5	95.6	85.2	75.3	70.3	56.1	41.9	74.7	64.9	79.9	30.5	41.7	72.7